



Lebanese University



Faculty of Sciences

MASTER THESIS

In order to obtain the

PROFESSIONAL MASTER

Biodiversity: Management and Conservation of Natural Resources

Issued by:

Faculty of Sciences of the Lebanese University

Presented and defended by:

Elias MATAR

on

Title

Climate Change Impacts on Agricultural Systems in Lebanon

Supervisor

Marie-Thérèse Abi-Saab, PhD

Examiners

Jean Michael Stephan, PhD

Charbel Mouawad Mouawad, PhD

This work was prepared at the Lebanese Agricultural Research Institute (LARI).

Table of Contents

Acknowledgements	4
Abstract.....	4
Résumé	5
Introduction	6
Chapter I: Literature review.....	7
1 Introduction to climate change	7
2 Climate change and agriculture	8
2.1 Impact of climate change on crop yield	8
2.2 Agricultural adaptation measures to climate change	9
2.3 Climate change in Lebanese agriculture.....	9
3 Use of modeling approaches for simulating the climate change impacts on agriculture....	11
Collected examples from different regions	12
3.1 The crop growth model AquaCrop for yield predictions under climate change	14
4 Climate change mitigation	16
Chapter II: Methodology.....	17
1 Study area	17
2 Climate projections.....	17
3 Soil database.....	20
4 AquaCrop: Crop water productivity model	20
5 Wheat simulations	20
6 Data Analysis.....	21
Chapter III: Results.....	21
1 Future change in Lebanon's climate.....	21
1.1 Change in Rain	21
1.2 Change in temperature.....	24
2 Impact of climate change on wheat yield.....	29
2.1 Rainfed wheat.....	29
2.2. Irrigated wheat	31
3 Impact of climate change on wheat water productivity	34
Chapter IV: Discussion	37
Chapter V: Conclusion and future work	38
References	38
LIST OF ABBREVIATIONS	41
LIST OF SYMBOLS	42

LIST OF FIGURES.....	42
LIST OF TABLES.....	43

Acknowledgements

I would like to thank my supervisor, Dr. Marie-Thérèse Abi-Saab, for her guidance throughout this thesis, as well as the whole jury board: Dr. Jean for his help in correcting the manuscript, and Dr. Hassane, General Director of the Master's program, for helping me defend my thesis on time. I would also like to thank my family and friends, who stood by me and supported me throughout my studies.

Abstract

Lebanon is experiencing rising temperatures and decreasing precipitation, which make the country more vulnerable to drought. In Lebanon, wheat makes up almost 66% of the cultivated cereal crops, making it an important crop. The aim of this study was to assess the changing climate factors and their impact on wheat yield using AquaCrop. Two future periods were evaluated, mid-century (2046-2065) and end-century (2081-2100), and compared to the reference period (1986-2005) under two projections, RCP4.5 and RCP8.5. Precipitation is projected to decline, on average, by 8% (RCP4.5) to 15% (RCP8.5) at mid-century and by 14% (RCP4.5) to 30% (RCP8.5) at end-century. Minimum and maximum temperatures keep rising throughout, reaching by end-century average increases of 2.10 °C (RCP4.5) to 4.31 °C (RCP8.5) in minimum temperature and 2.35 °C (RCP4.5) to 4.72 °C (RCP8.5) in maximum temperature. Yield is expected to increase under both rainfed and irrigated conditions in several agroclimatic zones, with the highest increases in Mount Lebanon and the South. In other regions yield is expected to decrease, most noticeably in Baalbek-Hermel because of crop failure. Under rainfed conditions, wheat water productivity is expected to increase in most AHZ under all scenarios, but to decrease in areas such as Baalbek-Hermel, where rainfall is too scarce to trigger planting and yield accumulation. Under irrigated conditions, water productivity increases where rainfall declines and crops rely more on irrigation. These results suggest that a multi-scale perspective is useful for assessing climate change impacts on agriculture, and that more adaptation strategies are needed to mitigate yield losses.

Résumé

Le Liban fait face à une hausse des températures et une baisse des précipitations, ce qui rend le pays plus vulnérable à la sécheresse. Au Liban, le blé représente près de 66 % des cultures céréalières, en faisant une culture importante. L'objectif de cette étude était d'évaluer l'évolution des facteurs climatiques et leur impact sur le rendement du blé à l'aide du modèle AquaCrop. Deux périodes futures ont été évaluées, le milieu du siècle (2046-2065) et la fin du siècle (2081-2100), et comparées à la période de référence (1986-2005), selon deux projections, RCP4.5 et RCP8.5. Les précipitations devraient diminuer en moyenne de 8 % (RCP4,5) à 15 % (RCP8,5) au milieu du siècle, et de 14 % (RCP4,5) à 30 % (RCP8,5) à la fin du siècle. Les températures minimales et maximales continuent d'augmenter sur toute la période, avec des hausses moyennes, à la fin du siècle, de 2,10 °C (RCP4,5) à 4,31 °C (RCP8,5) pour la température minimale et de 2,35 °C (RCP4,5) à 4,72 °C (RCP8,5) pour la température maximale. Le rendement devrait augmenter, en conditions pluviales comme irriguées, dans plusieurs zones agroclimatiques, avec les plus fortes hausses au Mont-Liban et au Sud. Dans d'autres régions, le rendement devrait diminuer, le plus nettement à Baalbek-Hermel en raison de l'échec des cultures. En conditions pluviales, la productivité de l'eau devrait augmenter dans la plupart des zones, mais diminuer à Baalbek-Hermel, où les précipitations sont trop faibles pour déclencher le semis et la formation du rendement. En conditions irriguées, la productivité de l'eau augmente là où les précipitations baissent et où les cultures dépendent davantage de l'irrigation. Ces résultats suggèrent qu'une perspective multi-échelle est utile pour évaluer les impacts du changement climatique sur l'agriculture, et que davantage de stratégies d'adaptation sont nécessaires pour atténuer les pertes de rendement.

Introduction

Worldwide, climate change constitutes a real challenge for food security. Particularly, the rising trends of temperature and CO₂ concentrations as well as the changing rates of precipitation, evaporation and ground water levels would affect the availability of water resources and therefore food production (IPCC et al., 2021).

The impact of climate change on crop yields could be negative but also positive. In fact, an increase in atmospheric CO₂ could induce carbon fertilization that may help increase yields of C₃ crops such as barley, wheat, soybeans, and sugar beet. However, some plants could not be affected by this phenomenon, such as maize and sunflower, and therefore, climate change would lead to a stagnation or decrease of their yields, especially under rainfed conditions. This is the case of C₄ plants that can no more increase their CO₂ assimilation when the concentration of this gas rises in the atmosphere (Blanco et al., 2017).

Lebanon is experiencing climate change through increasing temperature and decreasing precipitation trends which make the country more vulnerable to drought risks (Verner et al., 2018). In Lebanon, wheat makes up almost 66% of the cultivated cereal crops. (MoA & FAO, 2010). It is grown under rainfed but also under irrigated conditions. Nowadays, wheat yield fluctuates mainly due to scarce and erratic precipitation, high temperature variations, and a non-rational use of supplemental irrigation (Abi Saab, Houssemeddine Sellami, et al., 2019; Abi Saab, Jomaa, et al., 2019). Indeed, applying crop growth models may contribute to understand the future of wheat yields in Lebanon (Abi Saab et al., 2021).

Crop growth models such as AquaCrop have been used to project climate change impacts on wheat in several Mediterranean and semi-arid regions, including Sardinia and Tunisia (Bird et al., 2016) and the Banas River Basin in India (Dubey & Sharma, 2018). For Lebanon, AquaCrop has been calibrated and shown to predict the attainable wheat yield well, but only at the field scale (Abi Saab et al., 2021). To date, no study has applied the model across all of Lebanon's agro-homogeneous zones under downscaled climate projections to produce a national, spatially resolved assessment of how wheat yield and water productivity would change in the future. The present study addresses that gap.

The aim of this study was to investigate how the changing of Lebanon's climate during the 21st century would affect wheat growing in comparison to a reference period (1986-2005). The specific objectives were to simulate wheat yield and water productivity under rainfed and irrigated conditions using the crop simulating software AquaCrop that was developed by The Food and Agricultural Organization (Abi Saab et al., 2021). The simulations considered two emission scenarios, RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end of the century (2081-2100).

Chapter I: Literature review

1 Introduction to climate change

Climate change is the change of average weather patterns defining earth's global, regional and local climates on the long-term (IPCC, 2014). Global temperature has risen 1.18°C since the late 19th century. 2020 was the second warmest year on record according to NOAA's^[1] temperature data, being 0.98°C warmer than the mid-20th century (1951-1980) average and 1.19°C warmer than the pre-industrial period (Lindsey & Dahlman, 2021). Rising temperatures are not expected to decrease anytime soon, and surface temperature mean is expected to rise anywhere between 0.5°C to 2.6°C between 2041-2065 and up to 4.8°C by the end of the century rising from a 32°C to 39°C of average global temperature (Figure 1) (IPCC, 2014).

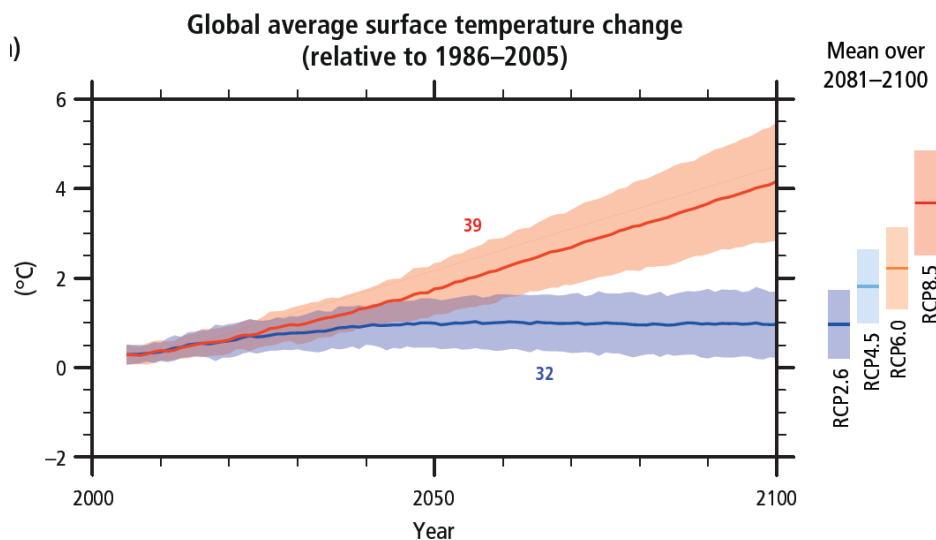


Figure 1 Global average surface temperature change

The main cause of global warming is the greenhouse gas emissions caused mostly by anthropogenic activity. Current concentrations of GHGs are the highest they've been in the last 800,000 years. As of April 2021, CO₂ concentrations have reached 416ppm, which is a 48% increase since 1750, the beginning of the industrial revolution (NASA, 2021). Methane levels have increased 172% from 694ppb in 1970 to 1892ppb as of December of 2020 (2° Institute, 2020a), and N₂O levels 23% from 271 to 334ppb (2° Institute, 2020b).

Climate change doesn't only include temperature rise: the top 100 meters of the ocean have increased of 0.33°C since 1969, the Greenland and Antarctic ice sheets have decreased in mass and According to NASA's Gravity Recovery and Climate Experiment, Greenland has been losing an average of 279 billion tons of ice per year between 1993 and 2019, while Antarctica has been losing around 148 billion tons of ice per year. Glaciers have been retreating, and extreme climatic events, such as hurricanes and intense rainfalls, have been on the rise (NASA, 2019).

In terms of rainfall, between 1951 and 1995 there has been a 20% decrease in precipitations in Mediterranean regions (Piervitali & Colacino, 2003), further decrease in precipitations and is expected in the future, and an increase in mean temperature and extreme weather events such as droughts and floods (Abd-Elmabod et al., 2020; Bird et al., 2016) and parts of these regions are at risk of desertification especially the southern ones by 2030 (Piervitali & Colacino, 2003). Precipitations are expected to increase up to 50% on the poles and the equator due to melting of sea ice, and. In the Andalusian region, models suggest an increase in precipitations by 2040, however the overall trend in the years 2070-2100 is lower precipitation means (Abd-Elmabod et al., 2020). A 30-40% decrease in rainfall events is expected by the end of the century in arid regions such as Botswana (semi-arid climate) with the strongest impact on dry winter seasons (Batisani & Yarnal, 2010). In tropical regions, rainfall is expected to increase however the number of rainy days is expected to decrease in the future (IPCC, 2014). Climate change also represents a threat to soil properties and capabilities including water availability, carbon storage and yields (Abd-Elmabod et al., 2020).

2 Climate change and agriculture

Agricultural crops and farming methods are selected to be suitable for the climate of a specific region making agriculture a climate dependent bio-industry (Kim, 2008). Shifting in agricultural climate resources can have a double impact: the first on the hydrology sector with underground water, river flows and water quality of lakes and marshes which could cause a real threat for the irrigation of crops, and on the arable/livestock sector by affecting agricultural production and crop yield, changes in ecosystems like the emergence of blights and pests, migration of populations and changes and drifts of biodiversity (Kim, 2008). Changes in the climate system such as precipitation changes and the lengthening of frost-free season since 1980, have had a perceivable impact on the agriculture and crop development (NASA, 2019). Therefore, the main impact of climate change on agriculture could be divided into 3 categories: Impact on crop yield, impact on food security, and impact on the economy.

Change in water resources will affect food production through changing rates of precipitation and evaporation, ground water levels, and dissolved oxygen content (IPCC et al., 2021). Greenhouse gas concentrations, radiation, water scarcity, precipitation and temperature are the most impactful climate factors (Dubey & Sharma, 2018).

2.1 Impact of climate change on crop yield

Climatic water balance is another climate change parameter as a study performed in Romania about the past climate impact on crop yield between the years 1990-2013, found that there was a strong correlation between climatic water balance and crop yield in 21 counties out of 41 (51%) with an average dependence of 35% between water balance and crop yield. The same study also finds the importance of the slope parameter on loss of yield with a 1mm decrease of precipitation, and continue to state that a 100mm decrease in the annual precipitation means could cause a decrease of 1t/ha/yr in the future. Besides crop

water balance, evapotranspiration is also a climate change parameter that proved to have significant effects on crop yields (Právělie et al., 2020). A study on US great plains suggests that irrigated crops are less vulnerable to changing climate than non-irrigated crops (Kukal & Irmak, 2018). The impact is also a function of the growing stage, as maize yield seems to be unaffected by drought stress after the milking stage (Zhu et al., 2021). In Tunisia, spring droughts can cause more than a 20% reduction in cereal yields (Bahri et al., 2019). In Ethiopia, people rely heavily on rain for irrigation as they have 2 rainy seasons out of three in the rift Valley making water resources very valuable to them (Muluneh, 2020).

Weather extremes caused by climate change, such as drought or extreme heat can be the cause of an average 10% loss in cereal yield productions. During 1964-2007 it is estimated that 1,820 million t of cereals were lost due to extreme drought events and 1190 million tons due to extreme weather disasters (Lesk et al., 2016). Lesk also estimated a 6-7% of yield sensitivity with every 1°C of warming and developed countries seem to be more vulnerable than developing one due mainly to the difference between crop diversity (more diverse in low income countries), the use of risk-minimizing strategies in low-income countries, and lower fair-weather yields in low-income countries.

Vulnerability of crop yield to climate change is also linked to the type of soils as some soils are more vulnerable than others. Rainfed wheat could be facing 64% less yield in 2040-2070 on clay loam soils in Mediterranean regions (Bird et al., 2016). Yield variability is expected with changing climate in Europe, as soybean yield is expected to increase up to 12 % in some areas and decrease of 8% in others (Blanco et al., 2017).

2.2 Agricultural adaptation measures to climate change

In order to fight food insecurity, agricultural adaptation strategies are needed. Changes in land use can be an example of farmers' mitigation strategies. In Eij county in Iran between the years of 2005-2012 land use changed from planting fodder and seeds to pomegranate trees in multiple lands (Tahmasebi et al., 2020). Many findings confirm that market effects by counterbalancing yield changes with interregional adjustments of production and consumption are an important climate change mitigation strategy (Blanco et al., 2017).

In Lebanon, blackberries are being promoted as a new crop due to their tolerance to some drought and high aridity. Quinoa, native to south America is also being promoted as a market crop as it can well adapt to multiple climatic conditions and is highly nutritious (Verner et al., 2018).

2.3 Climate change in Lebanese agriculture

Due to its high topographic diversity, Lebanon has multiple different climatic zones, thereby multiple agroclimatic zones. 35% of Lebanon's surface is arable land formed from 360,000 hectares of which 102,471 hectares of temporary crops, 125,928 hectares of per-manent crops, and 3,801 hectares of greenhouses. 31% of Lebanese agriculture is represented by fruit trees, 23% olive trees, 20% cereals, 17% are vegetables and the rest are industrial crops such as tobacco and vineyards. Decrease in the amount of precipitation and higher

temperatures is threatening crops in Lebanon especially in the Bekaa valley, and by threatening cereal production, it may negatively impact food security. Cereals occupy 45000 hectares of the Lebanese territory with 65% of these lands being in the Bekaa valley, with wheat and Barley being the most important crops (Verner et al., 2018).

Since 1970, Lebanon has been experiencing a temperature rise of 0.3°C per decade. Precipitations have also been decreasing because of large-scale drivers such as radiative forcing from enhanced greenhouse effects. LARI reported a 40 percent rain decrease in 2017 than an average year, and the driest winter of 2013-2014 had only 431mm of rainfall which is a little more than half the normal average of 812mm. These climate change trends make the country even more vulnerable to drought risks (Verner et al., 2018).

Droughts can cause serious problems. They raise irrigation demands that are then fulfilled by groundwater abstraction. The increase in water pumping, the need to deepen the wells or drill new ones can be of great cost for all stakeholders. And blackouts that Lebanon faces on a daily basis causes pumps to fail multiple hours at a time (Verner et al., 2018).

Lebanon hasn't had much success in managing drought and the reason for that is its reliance on short-term emergency relief measures. The problem with this solution is that it doesn't look at droughts as a major threat for the country (Verner et al., 2018). That is why strategies to mitigate climate change and understand the impact of droughts are needed to face this issue and find ways to lower the damage caused by climate change.

Multiple challenges face Lebanon's drought risk management: droughts are currently managed by individual institutions through their own frameworks and not centralized. A national strategy for drought management is still missing due to lack of legislations and the absence of an emergency plan. Efforts to empower drought risk management on subnational level are very minimal, although some notable successes in drought management have been noticed where local municipalities had control over water resources for irrigation. Stakeholders have very little information about weather data as LARI and the meteorological department are the only institutions that provide weather reports without additional analyses and the latter is usually met with delays if information comes at all (Verner et al., 2018).

Euro-CORDEX climate model is a climate model that include future climate changes for Lebanon and is the best choice due to the small size of Lebanon, and the model having 0.11° (12km) resolution. Temperatures in Lebanon are expected to rise from 2.7 °C (RCP4.5) to 3.9 °C (RCP8.5) by mid-century, with less increase towards the coasts. Precipitation is also expected to decrease from 26 mm/year to 43 mm/year with the largest decreases adjacent to Lebanese mountains this change can negatively impact groundwater aquifers that depend on snowfall (Arab Center for the Studies of Arid Zones and Dry Lands (ACSAD) of the League of Arab States et al., 2019).

3 Use of modeling approaches for simulating the climate change impacts on agriculture

Crop vulnerability is considered as an important measure of resistance to a certain hazard, making it a key factor in risk assessment. Drawing vulnerability curves can be used in Agricultural research, which is where the modelling comes in. Multiple crop models are used to simulate crop yield losses, land use changes and economy in order to understand the impact of climate change on crop yields and risk assessment in order to mitigate climate change impact and be prepared for what's to come (Abd-Elmabod et al., 2020; Bird et al., 2016; Blanco et al., 2017; Dubey & Sharma, 2018; Zhu et al., 2021). Modeling approaches can also be used to explain the relationship between past climate data and yields (Bocchiola et al., 2019; Právělie et al., 2020). Data for simulations require precipitation, temperature, evapotranspiration and soil properties, which can be provided by GCMs and RCPs (Bocchiola et al., 2019; Zhu et al., 2021). Each GCM can have different properties (Table 1), HADGEM2 is a climate model that can be used to predict higher droughts in northern Europe than other climatic models such as the IPSI or MIROC. Analyzing simulations under multiple climate scenarios could help provide better projections as GCMs lack some evidence as to how accurate they can be (Blanco et al., 2017).

Climate models provided by the IPCC can reproduce with high confidence the observed continental-scale surface temperature patterns and multi-decadal trends, including the more rapid warming since the mid-20th century and the cooling immediately following large volcanic eruptions. These models however, perform less well with precipitation compared with surface temperature and the confidence in their representation of processes involving clouds and aerosols remains low (IPCC, 2014).

Radiative forcing is the downward minus upward RCPs or representative concentration pathways are scenarios that include a time series of all Greenhouse Gas emissions such as CO₂, as well as aerosols and chemically active gases in the atmosphere (IPCC, 2014). RCPs are currently divided into four scenarios, the first one being RCP 2.6 that is the pathway where radiative forcing peaks at approximately 3 W m⁻² before 2100 and then declines (the corresponding ECP assuming constant emissions after 2100). RCP4.5 and RCP6.0 are two intermediate stabilization pathways in which radiative forcing is stabilized at approximately 4.5 W m⁻² and 6.0 W m⁻² after 2100 (the corresponding ECPs assuming constant concentrations after 2150). RCP8.5 One high pathway for which radiative forcing reaches greater than 8.5 W m⁻² by 2100 and continues to rise for some amount of time (the corresponding ECP assuming constant emissions after 2100 and constant concentrations after 2250). Here are the changes in CO₂ concentrations according to different RCPs, which shows that in a worst case scenario we could be dealing with more than 150 GtCO₂/yr (IPCC, 2014).

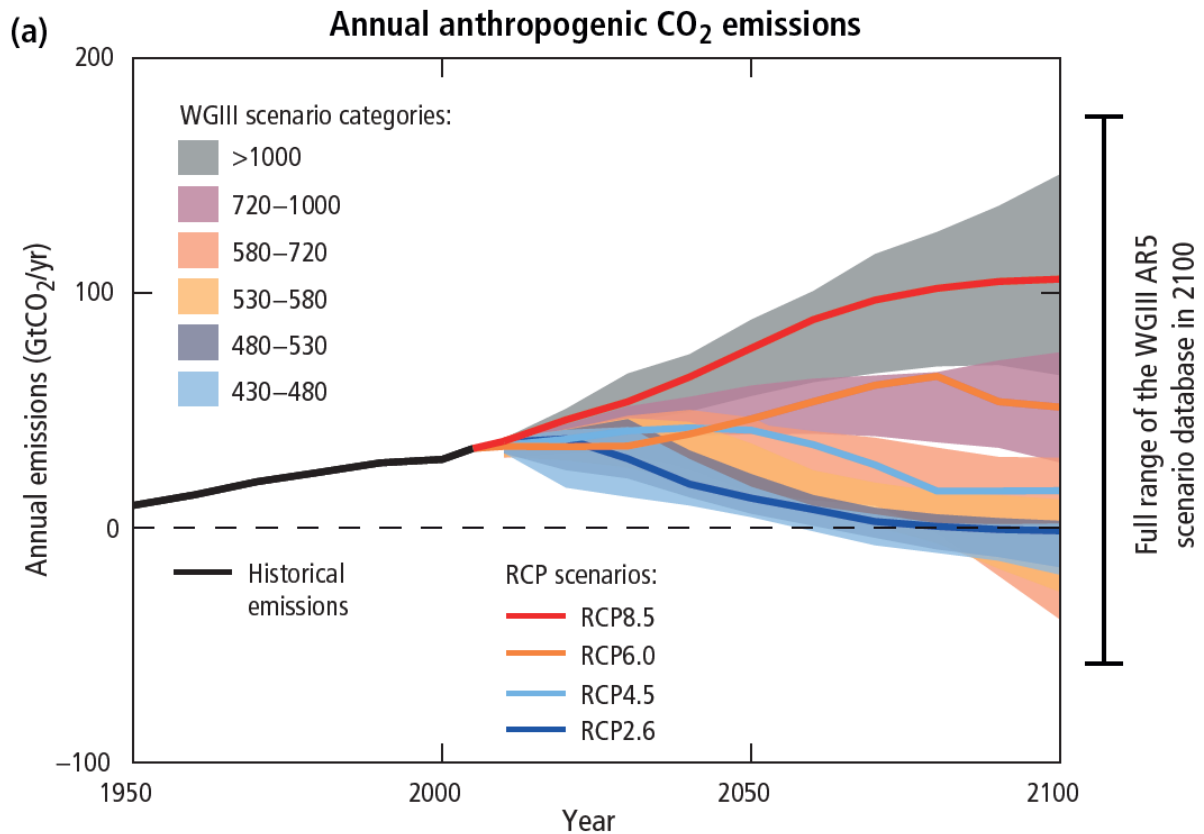


Figure 2 Annual anthropogenic CO₂ emissions according to different RCPs

Collected examples from different regions

Authors	Country	Crop	Agricultural Model	Climate Model	CC Scenario	Production and crop yield
(Zhu et al., 2021)	China	Maize	AquaCrop: Meteo: daily precipitation, max and min temp. Soil: texture and organic content of soil. Crop:	N/A	- Rain-fed only: less precipitation s. 100% Irrigation.	decreased with increasing DHI from 0.2 to 0.6
(Blanco et al., 2017)	European union	Wheat Barley Maize Potato Rye Rice Rapeseed Sunflower Soybeans	WOFOST crop model for EU countries LPJmL for non-eu countries CAPRI for agro-economic results (global and eu specific)	RCP 8.5 w/m ² (GCMs), namely HadGEM2-ES MIROC-ESM-CHEM and IPSLCM5A-LR	- rising radiative forcing pathway leading to 8.5 W/m² in 2100. - Temperature changes. - Rainfall patterns variation (drier and wetter).	Carbon Fertilization: Increase in yield or stagnation. Decrease for sunflowers, maize and Potato. No carbon fertilizations: Decrease in all crop yields

(Bird et al., 2016)	Sardinia	Wheat	AquaCrop EToCalc		- Planting density. - Sowing date. - Number of days for crop maturity.	2040-2070 time period: -Clay loams: 64% decrease, 15% crop failure. -Sandy loam: increase 8% -Sandy clay loam: 26% increase
	Tunisia	Tomato				Decrease on Sandy clay loams of 81% If tomatoes are sown on sandy loams, increase of 12%
(Dubey & Sharma, 2018)	Banas River Basin, India	Wheat, Barley, Maize	AquaCrop	CORDEX-SA(CNRM-CM5, CCSM4 and MPI-ESM-LR) for RCP 4.5 and RCP 8.5	Lower precipitation (RCP 4.5:-20mm, RCP 8.5: -48mm) Higher CO ₂ (RCP 4.5: 449.43, RCP 8.5:473.443 ppm)	Wheat and barley: less increase in 8.5 than 4.5 Maize: higher increase in 8.5
(Abd-Elmabod et al., 2020)	Andalucia, southern Spain.	Wheat Sunflower	MicroLEIS (Terraiza, Cervatana) evaluate soil productivity as bioclimate deficiency/yield reduction, and general land suitability		A1B climate scenario Lower precipitation Higher temperature	Aridity index increasing, humidity index & precipitation index decreasing Decrease in sunflower yield ranged from 10% to 80%. Wheat decrease was negligible. Water surplus decreased and the water deficit increased in all soil units for all future years
(Warnatzsch & Reay, 2020)	Malawi	Maize	AquaCrop	RCMs produced by different organisations within the Coordinated Regional Climate Downscaling Experiment (CORDEX).	RCP 4.5 and 8.5	Fast-development cultivar shows a smaller decrease than slow-development under minimum precipitation. Sowing date has a significant

						impact on cultivar
(Bocchiola et al., 2019)	Nepal	Wheat Rice Maize	Poly-crop	CCSM4 ECEarth ECHAM6 RCP (2.5,4.5,8.5)	9 CC scenarios: CCSM4 + RCP (2.5,4.5,8.5) ECEarth + RCP (2.5,4.5,8.5) ECHAM6 + RCP (2.5,4.5,8.5)	Increase in Maize and Wheat yield with altitude until 2050, decrease between 2050-2100 due to population growth stabilization. Warmer fall-winter could potentially increase wheat yield.

Table 1 Results from different crop simulation models

3.1 The crop growth model AquaCrop for yield predictions under climate change

3.1.1 Model Description

AquaCrop is a dynamic crop growth model that uses mathematical calculations to calculate yield response to water of herbaceous crops, in other words, it measures the yield quantity as a function of water consumption under rainfed, supplemental, deficit, and full irrigation conditions that are shown by the equation:

$$\frac{(Y_x - Y_0)}{Y_x} = k_y \left(\frac{ET_x - ET_0}{ET_x} \right)$$

where Y_x and Y_0 are the potential and the actual crop yields (kg/m²), ET_x and ET_0 are the potential and actual evapotranspiration (mm), respectively, and k_y is the conversion coefficient between crop yield and water stress.

The core equation that AquaCrop is based on is:

$$B = WP \times \sum Tr$$

With:

B is the biomass produced cumulatively (kg per m²),

Tr is the crop transpiration (either mm or m³ per unit surface), with the summation over the time period in which the biomass is produced,

WP is the water productivity parameter (either kg of biomass per m² and per mm, or kg of biomass per m³ of water transpired).

To compute transpiration and distinguish soil evaporation from transpiration, the model employs canopy ground cover rather than the leaf area index (LAI). The product of biomass and harvest index is used to compute crop yield (HI). HI increases linearly with time after a lag phase, at the start of yield formation period, until near physiological maturity. Other

than for the yield, there is no biomass partitioning into the various organs. Crop responses to water deficits are simulated with four modifiers that are functions of fractional available soil water modulated by evaporative demand, based on the differential sensitivity to water stress of four key plant processes: canopy expansion, stomatal control of transpiration, canopy senescence, and HI. The HI can be modified negatively or positively, depending on stress level, timing, and canopy duration. The model is aimed mainly at practitioner-type end-users such as those working for extension services, consulting engineers, governmental agencies, nongovernmental organizations, and various kinds of farmers associations. It is also designed to fit the need of economists and policy specialists who use simple models for planning and scenario analysis. (Steduto et al., 2012). AquaCrop is the most suitable crop model to measure the relationship between crop growth and water balance (Zhu et al., 2021).

3.1.2 Examples of Aquacrop use for climate change simulations

Authors	Country	Crop	Agricultural Model	Climate Model	CC Scenario	Production and crop yield
(Muluneh, 2020)	Rift Valley, Ethiopia	Maize	AquaCrop	ECHAM5 BCCR_BCM2.0 CNRM-CM3 CSIRO-Mk3.5 INM-CM3_0 MIROC3.2	Increase in temperature: 2021-2050: +1.16°C Tmax +1.86°C Tmin 11% less precipitation 2066-2095: +3.16Tmax +3.95Tmin 8% less precipitation	2021-2050: -8.8% yield loss 2066-2095: -9.4% yield loss
(Warnatzsch & Reay, 2020)	Malawi	Maize	AquaCrop	RCMs produced by different organisations within the Coordinated Regional Climate Downscaling Experiment (CORDEX).	RCP 4.5 and 8.5	Fast-development cultivar shows a smaller decrease than slow-development under minimum precipitation.
(Voloudakis et al., 2015)	Greece	Cotton	AquaCrop	A1B climate scenario HadRM3, C4I, REMO-MPI, ETHZ, CNRM, DMI-HIRHAM, KNMI, SMHI,	+1.8°C, -11% precipitation by 2050 +4°C, -24% by 2100	According to climate models, there could be an increase for cotton yields or a decrease. Example of Alexandroupoli had a -138% decrease to +77% increase between 2021-2050

(Hassanli et al., 2016)	Karaj region, Iran	Maize	AquaCrop SALTMED SWAP			Decrease in crop yield with the increase of irrigation water salinity
-------------------------	--------------------	-------	-----------------------	--	--	---

Table 2 Example of AquaCrop use for climate change simulations

4 Climate change mitigation

Extreme droughts are expected to increase 29% during the 21st century, as well as their frequency and durations (Zhu et al., 2021). A study by (Kukal & Irmak, 2018) suggests that the Irrigated cropping systems are more resilient to climatic variability than non-irrigated counterparts because irrigation methods are an efficient drought-mitigation approach in terms of preserving grain production, but non-irrigated yields are significantly more sensitive to climate impacts.

LARI developed EWS (environmental weather systems) and weather forecasting facilities to that provide farmers with weather data. They also launched an application for weather data in 2015 reaching around 50,000 farmers to help farmers with choosing their sowing dates and additional information for sowing dates, irrigation efficiency information serves as a EWS for Lebanese farmers. sowing dates, irrigation efficiency information, and pest and disease management. The application also advises farmers on when to cultivate rainfed crops to avoid yield loss and promote sustainable use of water resources.

Between 25,000 and 40,000 hectares of cultivated rainfed wheat land are available. For the initial watering, the crop requires roughly five to ten millimeters of irrigated water. It takes between 100 and 500 cubic meters of water to irrigate a hectare of wheat, depending on a variety of conditions. As a result, using the EWS to manage irrigation can save millions of cubic meters of fresh water in Lebanon (Verner et al., 2018).

Aridity will continue to rise, as will the intensity and frequency of droughts, notably in central Lebanon, according to future climatic projections. These shifts are critical for Lebanese agriculture and the people who rely on it. Droughts in the agricultural sector can have social, economic, and environmental consequences (Verner et al., 2018).

Chapter II: Methodology

1 Study area

The whole Lebanon was considered as the study area. Lebanon's climate is changing (Verner et al., 2018). Lebanon is characterized by hot, dry summers and cool, moist winters. But the country has experienced an annual mean temperature increase of $\sim 0.3^{\circ}\text{C}$ per decade since about 1970.

There are also 360,000 hectares of arable agricultural lands in Lebanon. This comprises 35 percent of the country's surface area (MOA, 2010). The greatest concentration of agricultural lands is in the Bekaa Valley (42 percent of the total cultivated area), followed by northern Lebanon (26 percent), Southern Lebanon (22 percent), and Mount Lebanon (9 percent). These lands grow a wide variety of crops. Thirty-one percent of total agricultural production are fruit trees, 23 percent are olives, twenty percent are cereals, 17 percent are vegetables, and the remaining nine percent are industrial crops, like tobacco, grape vineyards, and others (MOA 2010). Cereal yields are vulnerable to decreased rainfall. The most vulnerable areas for cereal crops are in the Bekaa Valley where reduced precipitation is common during the growing season. Higher temperatures are also a factor for cereal production. In this country, the agricultural zones are provided according to the Agriculture Homogeneous Zoning (AHZ) method (FAO, 2006), as represented in Figure 3. The description of the different Agro-Homogeneous Zones of Lebanon is provided in Table 2.

2 Climate projections

The Earth System Grid Federation (ESGF) platform was used to download climatic data from the Coordinated Regional Climate Downscaling Experiment (CORDEX). Particularly, the European Domain (EURO-CORDEX), which extends beyond Europe to include North Africa and the Eastern Mediterranean, was used to provide climatic data at a spatial resolution of 0.11° (12.5 km). Compared to other domains such as the MENA-CORDEX domain which offers data at a coarser resolution (0.44°), EURO-CORDEX can offer a better assessment of climate change especially that Lebanon is of small size. In accordance with IPCC, two future time periods were evaluated: mid-century (2046-2065), and end-century (2081-2100) which were compared to the reference period (1986-2005), considering two projections: RCP4.5 and RCP8.5. According to the 5th IPCC report (2014), the former (RCP4.5) represents a moderate emissions scenario while the latter (RCP8.5) represents an extreme scenario. An ensemble was created on results driven by three GCM models: CNRM-CM5, EC-EARTH, and IPSL-CM5A-MR which were downscaled based on the Rossby Centre Regional Atmospheric Model (RCA4), developed by SMHI. This ensemble was also used and evaluated for Lebanon by (Arab Center for the Studies of Arid Zones and Dry Lands (ACSAD) of the League of Arab States et al., 2019).

Climatic data were downloaded as (.netcdf) files. Then, in QGIS, data representing each AHZ, was extracted and converted to excel file.

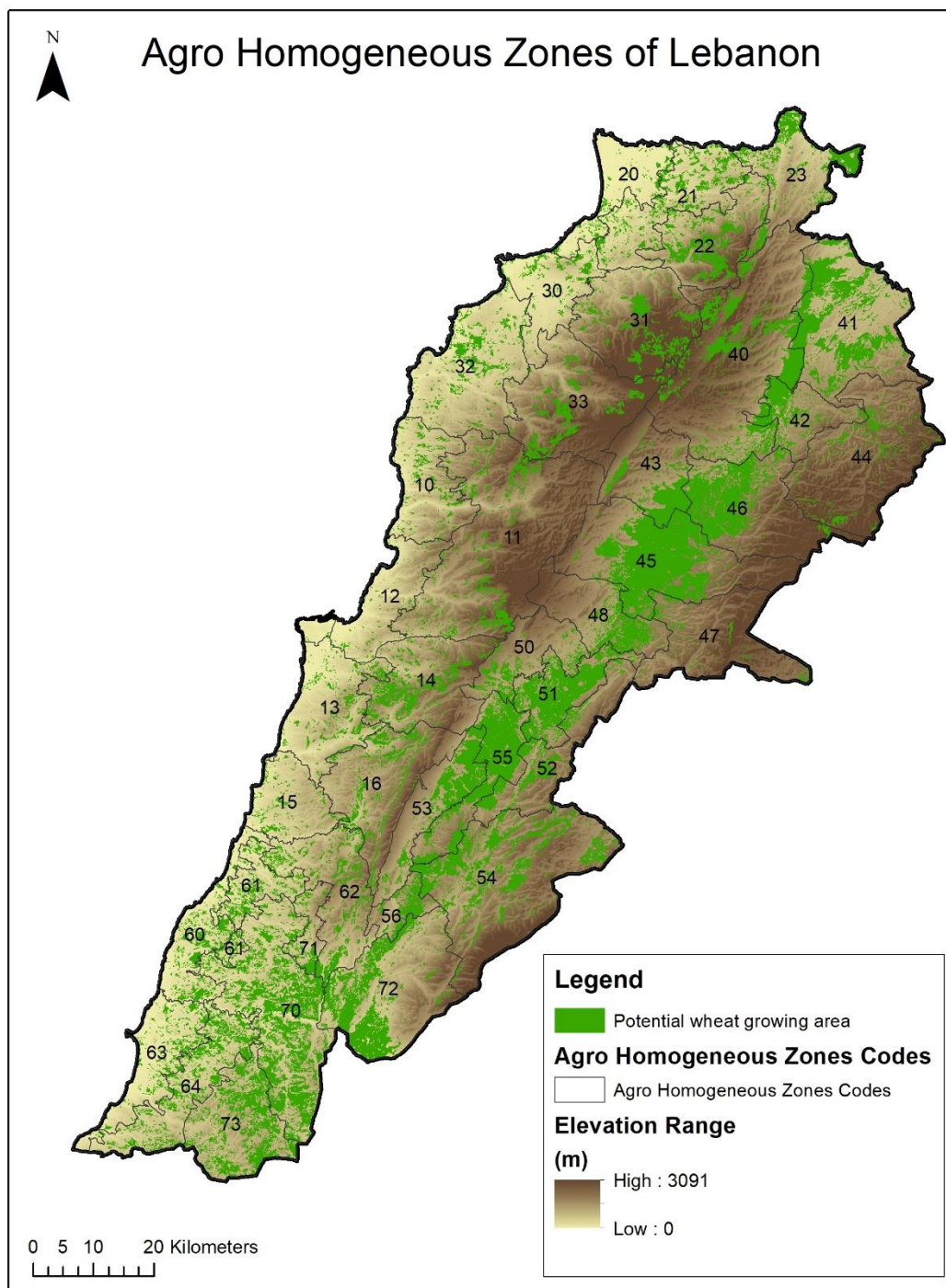


Figure 3 The Agro Homogeneous Zones of Lebanon and potential wheat growing area

Table 3 The description of the different Agro Homogeneous Zones of Lebanon

Governorate	Code	Description
North Lebanon	30	Olive zone in Tripoli
	31	Fruit trees zone of Danniyeh
	32	Olive zone in Batroun-Koura
	33	Mountainous zone of North Lebanon
	10	Coastal zone of Jbeil
Mount Lebanon	11	Mountainous zone of Northern Mount Lebanon
	12	Coastal zone of Northern Mount Lebanon
	13	Coastal zone of Central Mount Lebanon
	14	Mountainous of Central Mount Lebanon
	15	Iqlim El Kharroub
	16	Mountainous zone of Chouf
South Lebanon	60	Plain of Saida
	61	Plateau of Saida
	62	Jezzine
	63	Plain of Sour
	64	Plateau of Sour
Nabatiye	70	Nabatiye
	71	Iqlim El Teffah
	72	Marjeyoun/Hasbaya
	73	Bent Jbeil
Bekaa	50	Mountainous zone of Zahle
	51	Plain of Zahle
	52	Eastern watershed of Litani
	53	Western watershed of Litani
	54	Rachaya
	55	Plain of Litani
	56	Sohmor
Baalbek-Hermel	40	Western watershed of Orontes
	41	Northern plain of Orontes
	42	Southern plain of Orontes
	43	Deir El Ahmar
	44	Eastern watershed of Orontes
	45	Baalbek
	46	Chaat and Younine
	47	Britel
	48	Bednayel
Aakkar	20	Plain of Aakkar and Minieh
	21	Plateau of Aakkar
	22	Fruit trees zone in Aakkar
	23	Qobayyat zone

3 Soil database

The soil map of Lebanon, developed by the National Council for Scientific Research at a scale 1:50000 (Darwish et al., 2006), was intersected with the map of Lebanon AHZ to obtain the distribution of soils in each zone. For each region, the soil with the largest spatial distribution in arable areas was considered to be the dominant soil class. Accordingly, for each soil class, data on the soil type, horizon description and depth, and the physical characteristics (texture and stoniness) was taken from the reference booklet of the “Soil Map of Lebanon” (Darwish et al., 2006). Then after, collected data was processed through hydraulic properties calculator (Saxton et al., 1986) in order to estimate, for each soil class, the water contents at field capacity, saturation and wilting point and the saturated hydraulic conductivity.

4 AquaCrop: Crop water productivity model

In a previous study in Lebanon, (Abi Saab et al., 2021) showed that the AquaCrop model (Steduto et al., 2012) had good performance in predicting the attainable wheat yield. The AquaCrop model takes into account the impacts of water stress caused not only by stomatal closure but also by reductions in leaf expansion and premature canopy senescence. In addition, AquaCrop accounts for the effects of temperature on crop growth and the impacts of elevated atmospheric carbon dioxide concentrations on crop productivity, which makes it suitable for this study. Given the large spatial domain and the number of evaluations required, we used the open-source version of the FAO AquaCrop model (Foster et al., 2017). The AquaCrop model is capable of estimating crop productivity based on either fixed length of the crop cycle or on growing degree days (GDD). Simulation runs produce the daily outputs of several variables, such as yield (t ha^{-1}), biomass (t ha^{-1}), transpiration (mm), water productivity (kg m^{-3}), planting date, crop cycle (day) and irrigation (mm). In addition to the climate and soil data, AquaCrop requires information about the crop parameters that needs to be calibrated for use in field data, as well as the initial soil moisture conditions of the crop cycle. In addition, the model requires a time-series of CO_2 concentrations to run projections.

5 Wheat simulations

AquaCrop-calibrated crop parameters for wheat yield simulations in Lebanon were derived from the study of (Abi Saab et al., 2021). In addition, simulations used a time-series of CO_2 atmospheric concentrations extracted from the RCP4.5 and RCP8.5 IPCC scenarios. Therefore, wheat yield simulations conducted in this study include all physiological stresses related to elevated temperature, CO_2 concentration and water deficit. Bias-corrected maximum and minimum temperatures and the reference evapotranspiration and rainfall for both the baseline period and projections for the near future, middle and end century, were used as the input for the AquaCrop model. Planting dates for each simulation were not fixed but were determined by 5 consecutive days with at least 30 mm cumulative rainfall. This

approximation mimics the traditional management practices of smallholder farmers of the region that “wait for the rainfall to plant” rather than following a fixed calendar, and it allows for variations in the onset of the wet season under future climatic conditions. In addition, this fact provides enough initial moisture in the soil profile to ensure germination. Due to the high interannual rainfall variability, we assumed that there was complete crop failure whenever the amount of rainfall for triggering planting was not achieved by the end of December in each AHZ.

The simulations were carried out under rainfed conditions, then after, they were repeated under irrigated conditions by using the net calculation option available in AquaCrop, which quantifies the amount of water needed to avoid water stress effects. Thus, the method accounts for the amount of water needed maintain the soil profile at field capacity on a daily basis.

Finally, crop yield and productivity projections for the RCP4.5 and RCP8.5 scenarios for the periods 2046-2065 and 2081-2100 were compared to the baseline scenario 1986–2005. Results were grouped according to the different AHZ considered in Fig. 3.

6 Data Analysis

The output text files of the climate change simulations for each AHZ, time slice and emissions scenario were imported to separate worksheets in a Microsoft Excel 2010 workbook. Descriptive statistics, including percentiles, were then computed for the simulated yields for each time slice, emissions scenario and region using the Data Analysis tool. The same computation was done for the baseline yield data. Differences between projected yields and observed yields in the baseline period were calculated. Results were also shown on maps using GIS.

Chapter III: Results

1 Future change in Lebanon’s climate

1.1 Change in Rain

Precipitation is projected to generally decrease by mid-century, with decreases ranging from 57 mm/year (RCP4.5) to 114 mm/year (RCP8.5) on average (Figure 4). The largest decreases are apparent adjacent to the Lebanon Mountains, which can adversely impact groundwater aquifers dependent on snowfall. However, some rain increases were detected in selected areas located in the southern coast, mainly in the plain of Saida (2 mm/year), the plain of Sour (7 mm/year) and the plateau of Sour (44 mm/day). In addition, rain increases up to 4 mm/year were detected in the Bekaa valley of Lebanon, particularly in some AHZ of Baalbek-Hermel, but will have likely negligible impact. Other rain increases were also noted in plateau of Akkar (62 mm/year) and some areas of Mount Lebanon (ranging from 46 to 86 mm/year). By end-century, under RCP4.5, precipitation is projected to decrease across many areas, with an average of 103 mm/year (Figure 4). However, some areas will continue to have some rain increases such as the southern coast (11 mm/year in Plateau de Sour), the northern plain of Akkar (34 mm/year) and Mount Lebanon (13 mm/year Coastal zone of

northern mount Lebanon). By end-century, under RCP8.5, precipitation is projected to decrease across the entire study area, with an average of 218 mm/year. Similar to mid-century, decreases at end-century are more modest in the north, northeast, and the south, but intensify in the mid-section of the study area. These decreases which are based on EURO-CORDEX reflect an average decline of 8% (RCP4.5) to 15% (RCP8.5) at mid-century, and 14% (RCP4.5) to 30% (RCP8.5) at end-century.

The maps illustrating the future change in rain over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005) are provided in Figure 5.

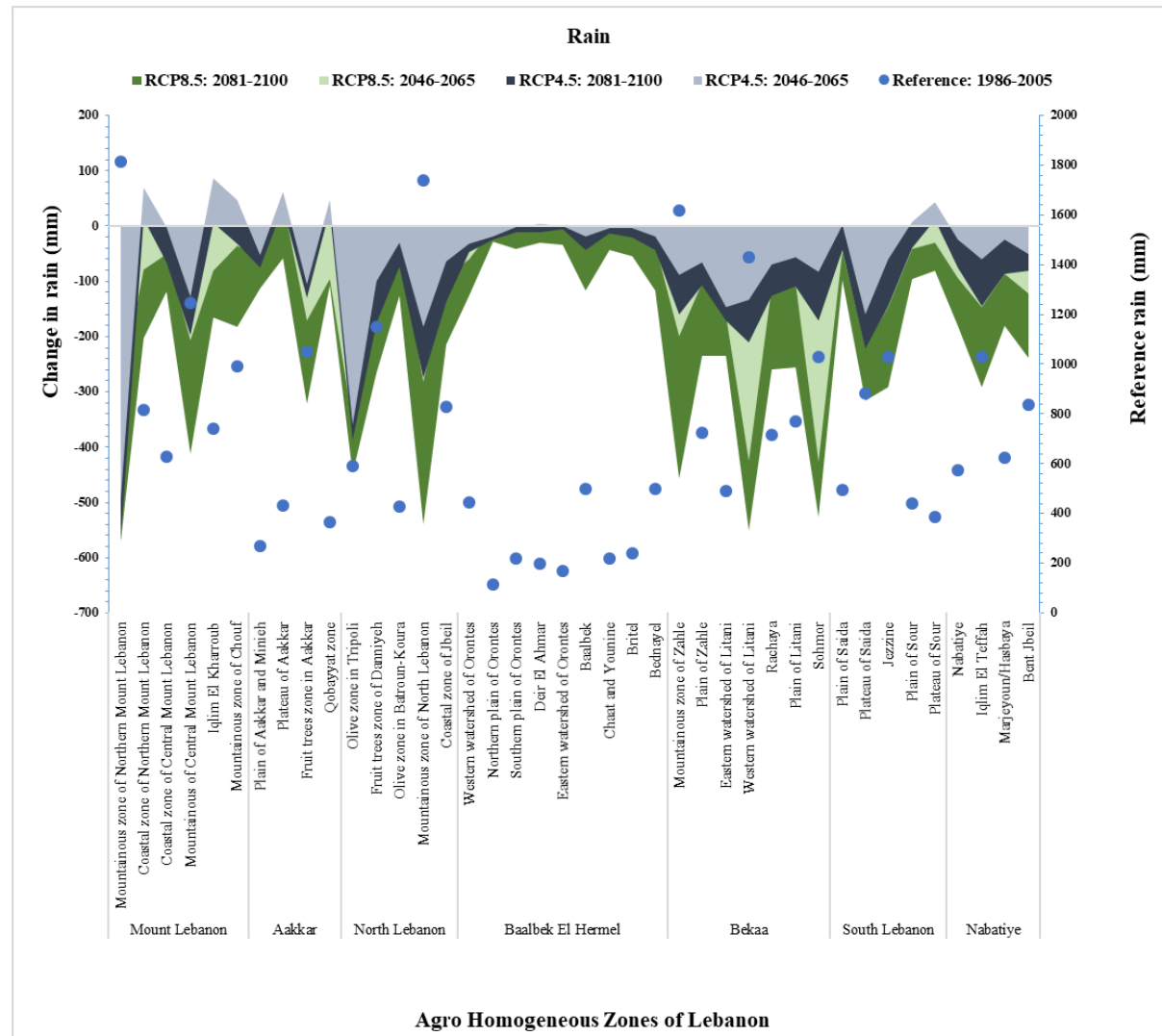


Figure 4 The average rain for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon

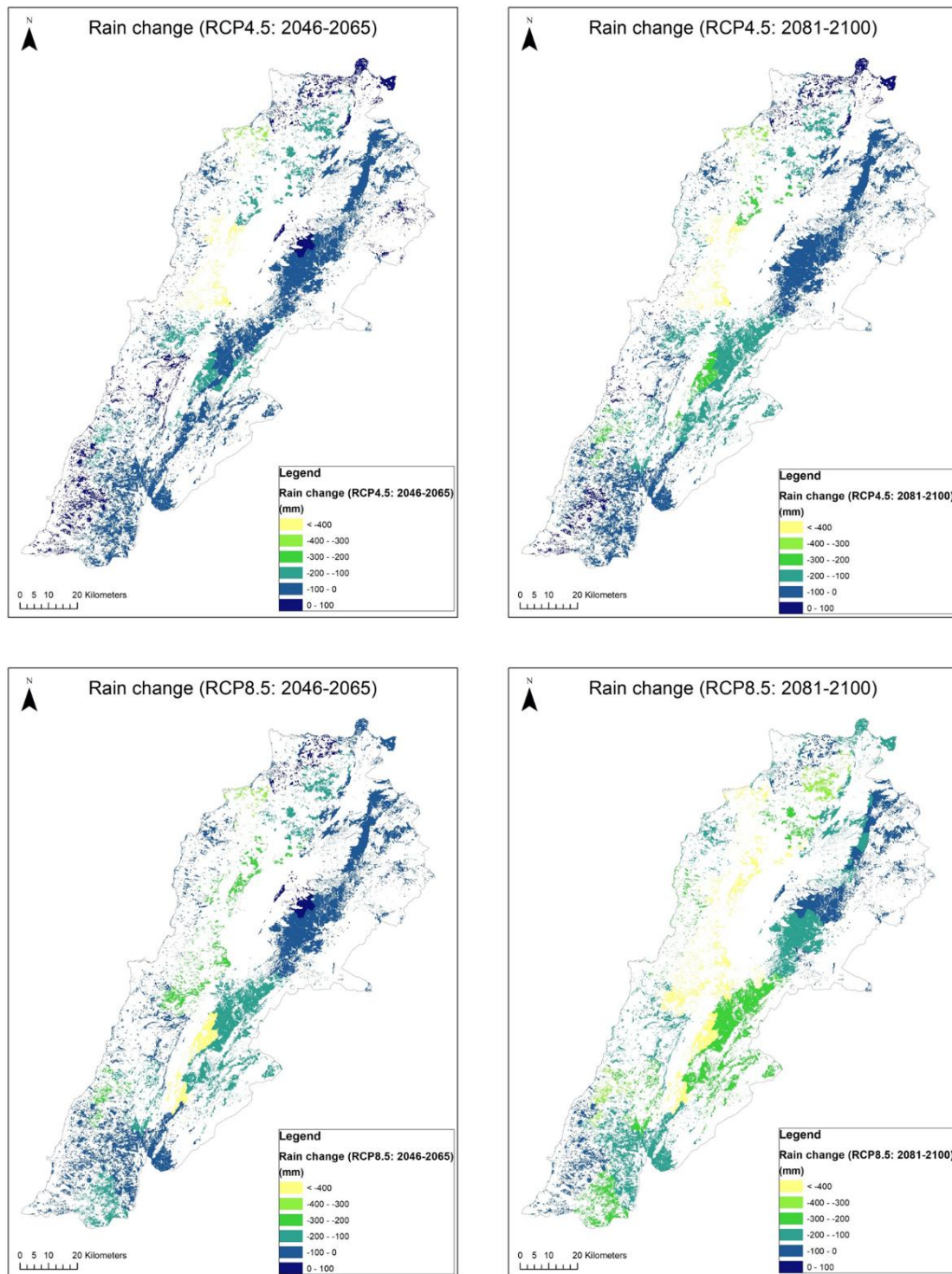


Figure 5 Maps of the future change in rain over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

1.2 Change in temperature

The projections at mid-century indicate that minimum temperatures in Lebanon are projected to rise by an average of 1.53°C (RCP4.5) to 2.30°C (RCP8.5), with smaller increases toward the coastal zone of northern Lebanon (Figure 6). The largest increases are projected in the northern Lebanon Mountains. At end-century, minimum temperature is projected to increase by an average of 2.10°C (RCP4.5) to 4.31°C (RCP8.5), when compared to the reference period (Figure 6). Similar to mid-century, the smaller increases are projected near the coast while the largest change is located in the northern Lebanon Mountains. For the maximum temperature, the projections at mid-century indicate that an average rise of 1.69°C (RCP4.5) to 2.53°C (RCP8.5) will occur (Figure 8). At end-century, maximum temperature is projected to increase by an average of 2.35°C (RCP4.5) to 4.72 °C (RCP8.5), when compared to the reference period (Figure 8).

Figures 7 and 9 show the maps of the future change in minimum and maximum temperatures over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

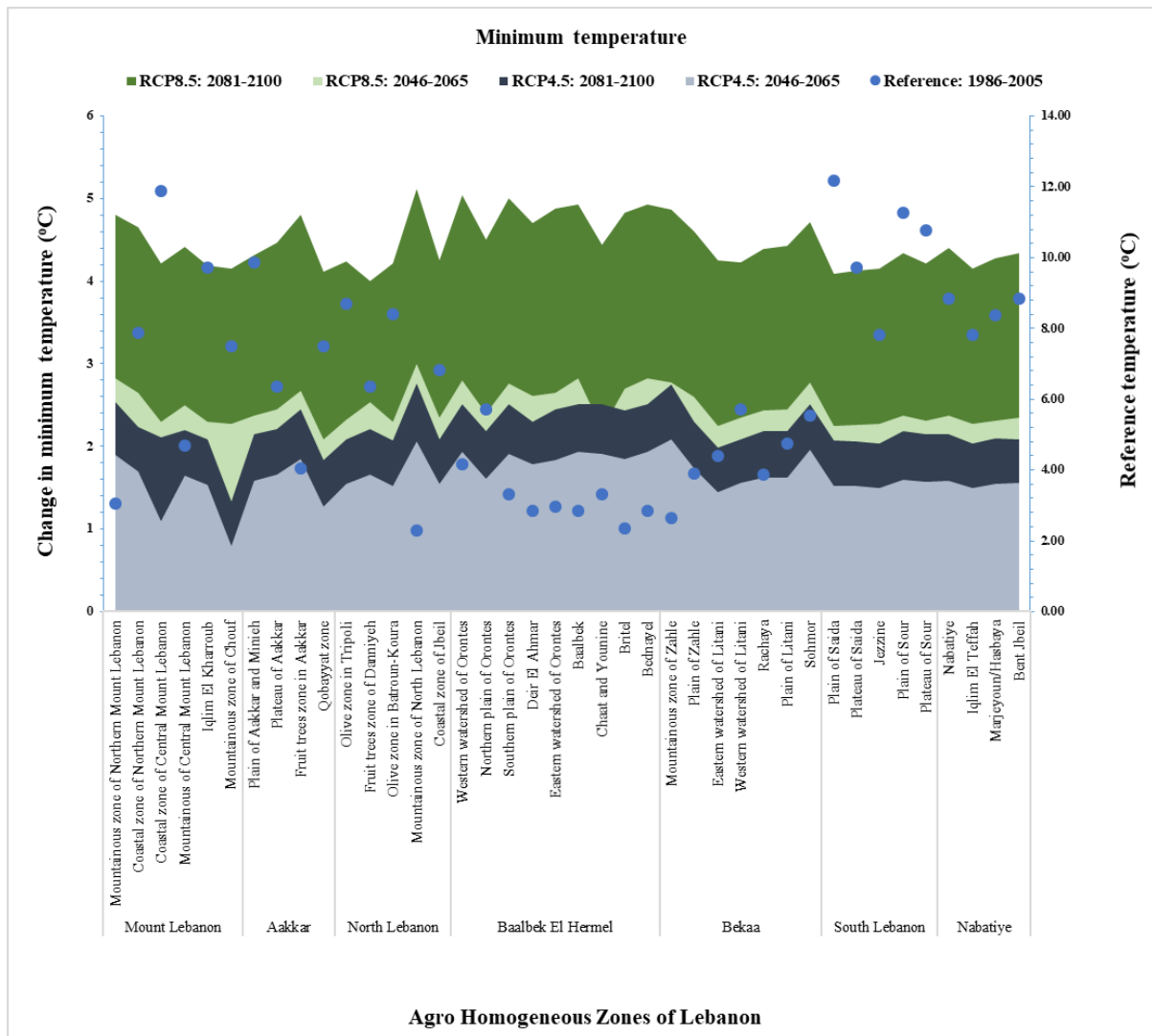


Figure 6 The average minimum temperature for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon.

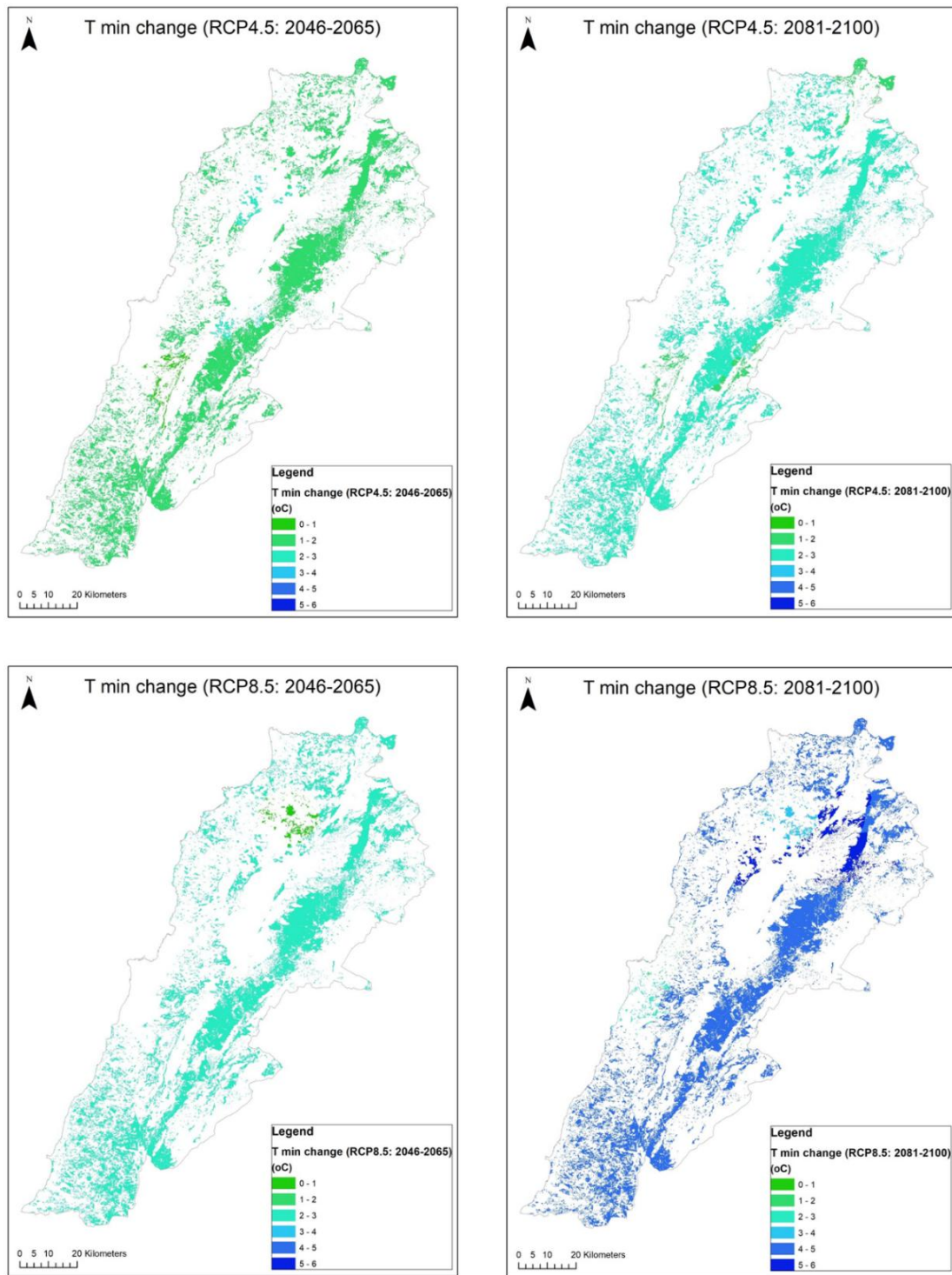


Figure 7 Maps of the future change in minimum temperature over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

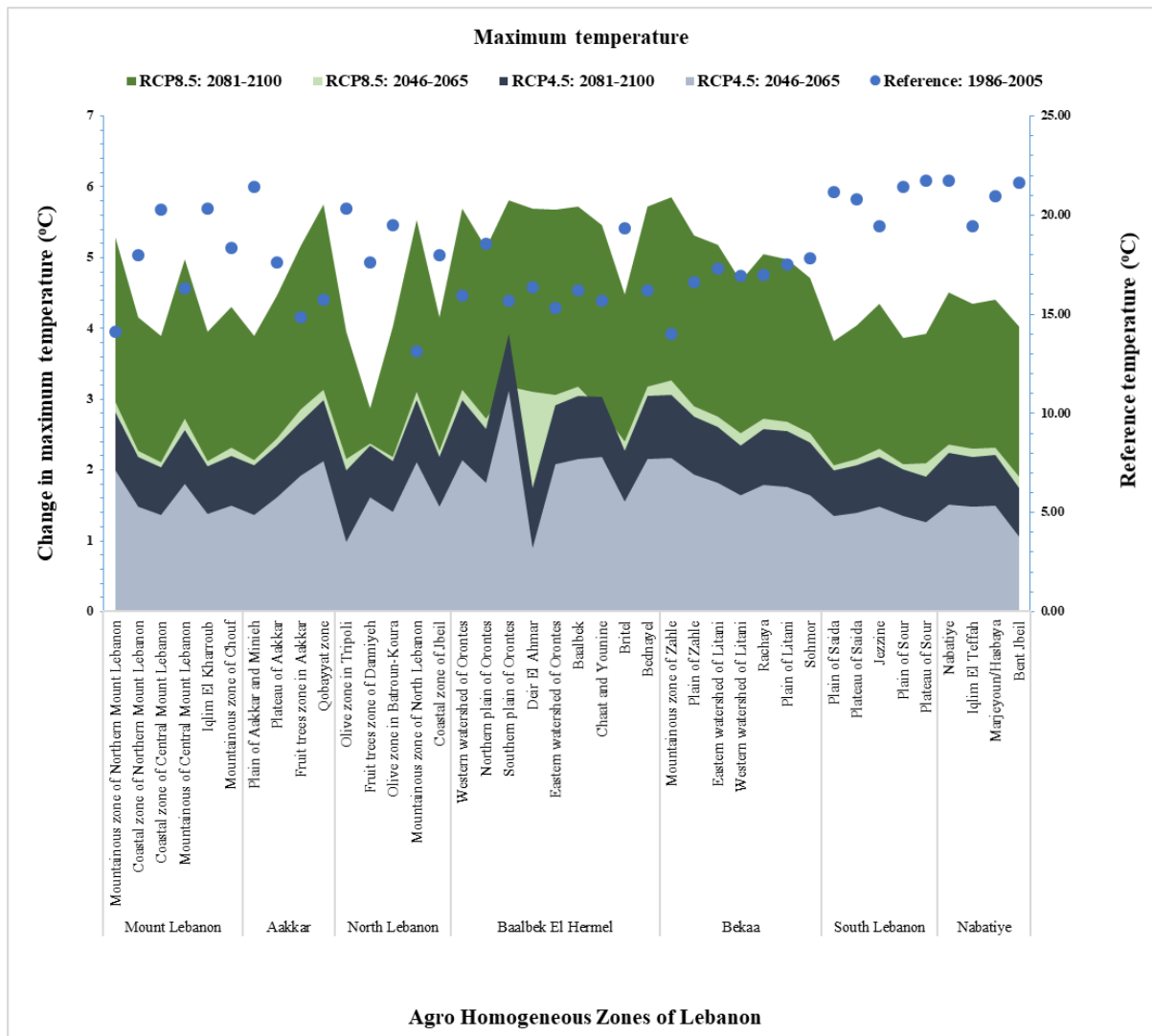


Figure 8 The average maximum temperature for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon

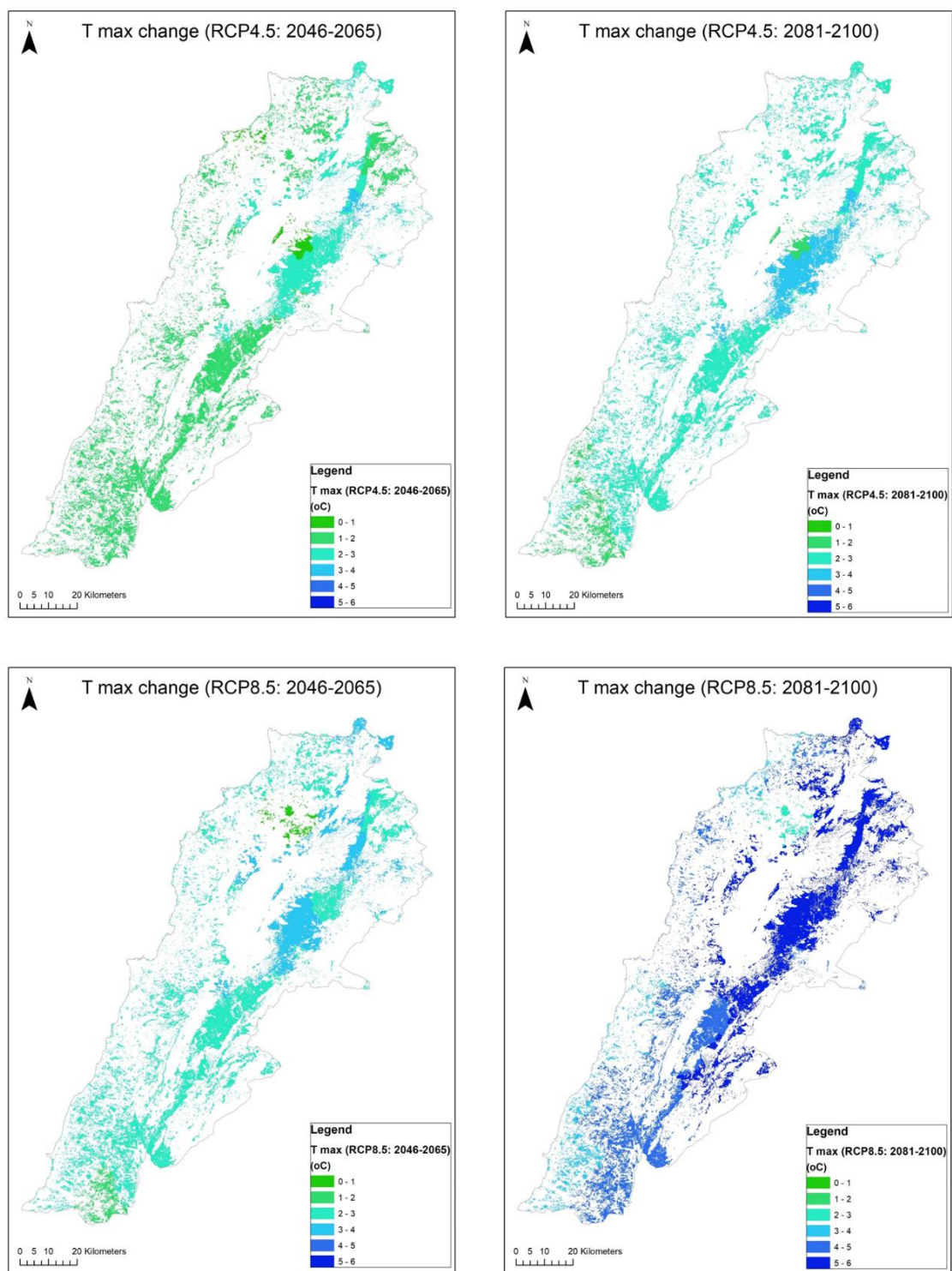


Figure 9 Maps of the future change in maximum temperature over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

2 Impact of climate change on wheat yield

2.1 Rainfed wheat

In regions where the amount of rain can trigger planting, there is no crop failure, so yield is expected to increase in the mid-century (2046-2065) especially in Mount Lebanon and South regions with a maximum of 1 t.ha⁻¹ and 6.5 t.ha⁻¹ under RCP4.5 and RCP8.5, respectively. However, the yield in the Baalbek-Hermel region is not expected to increase which could be due to the fact that the amount of rain is not enough to trigger planting before December. Therefore, crop failure would be induced for this region as shown in Figures 10 and 11.

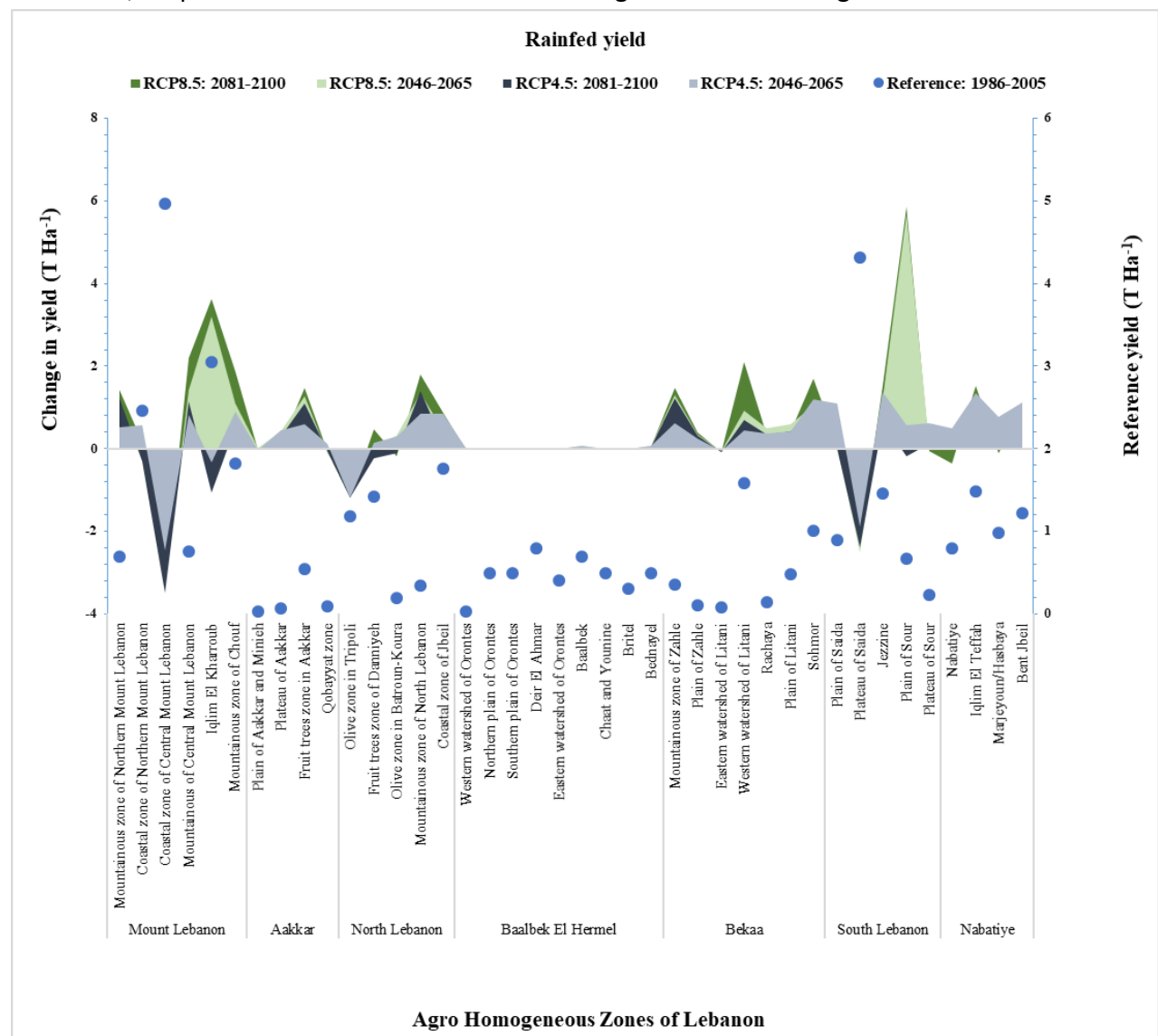


Figure 10 The average rainfed yield for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon.

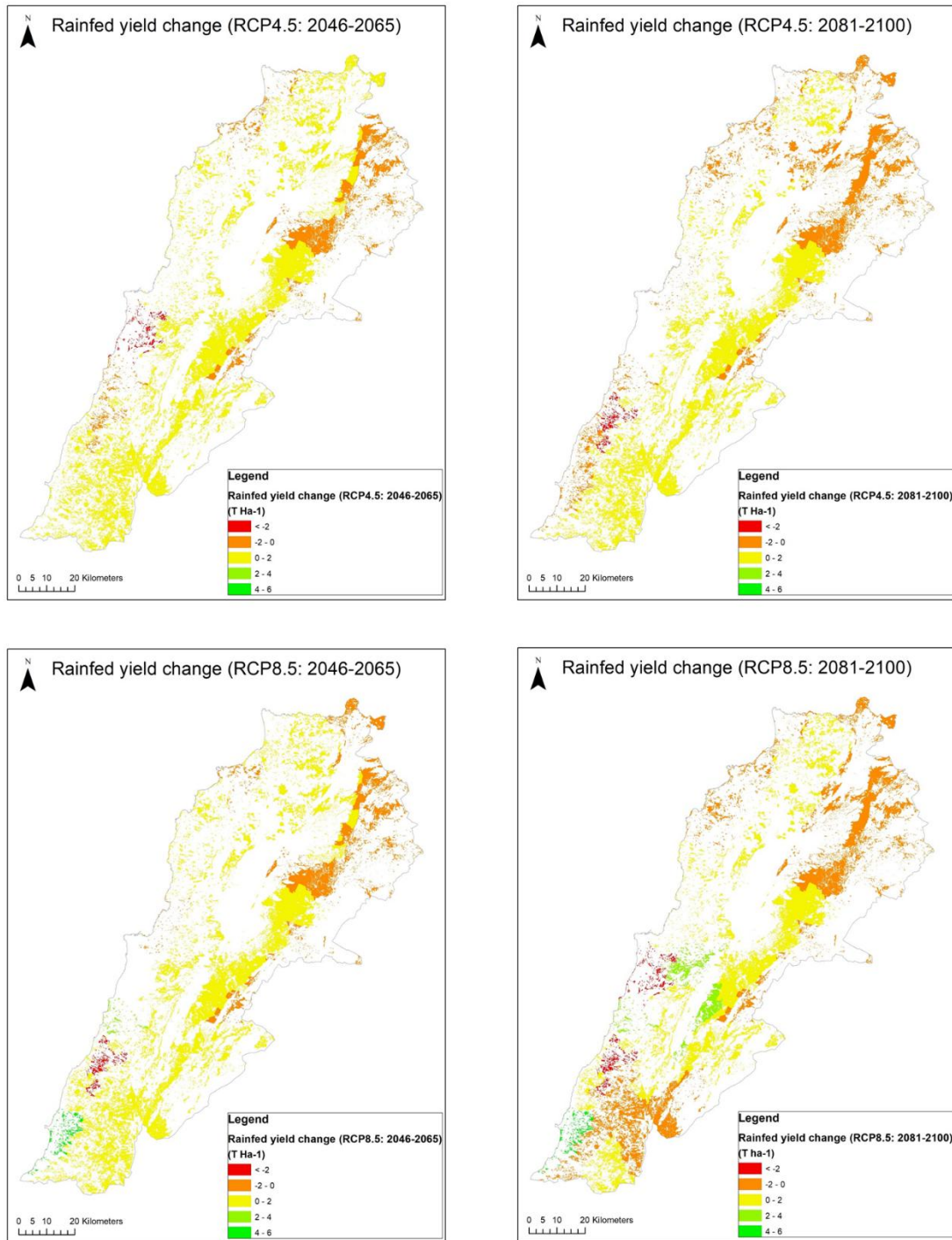


Figure 11 Maps of the future change in rainfed wheat yield under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

2.2. Irrigated wheat

To analyze whether the use of irrigation could minimize the loss of productivity due to climate change, Figures 12 and 13 show the same simulations as under rainfed conditions, except that irrigation was included. It is interesting to note that for the current climate, the use of irrigation can significantly increase the productivity of wheat, especially in the Bekaa valley. As expected, there was an increase in yield in the future in many AHZ due to the fact that irrigation suppresses the risk of loss related to dry spells. However, many AHZ are still projected to have a decrease in wheat yield even under irrigated conditions. Consequently, although the increase in CO₂ concentration typically leads to higher plant water-use efficiencies that positively influence wheat productivity, the shortening of the growing season is associated with higher air temperatures during the grain-filling period, which causes a reduction of yield. In addition, the decrease of precipitation can negatively affect the transpiration and biomass accumulation, further reducing crop yield. Based on the mutual interactions of those effects and considering the need to adapt to irreversible climate change effects, the identification of the most effective combination of strategies and technologies for each regional context remains the main challenge of the agriculture sector.

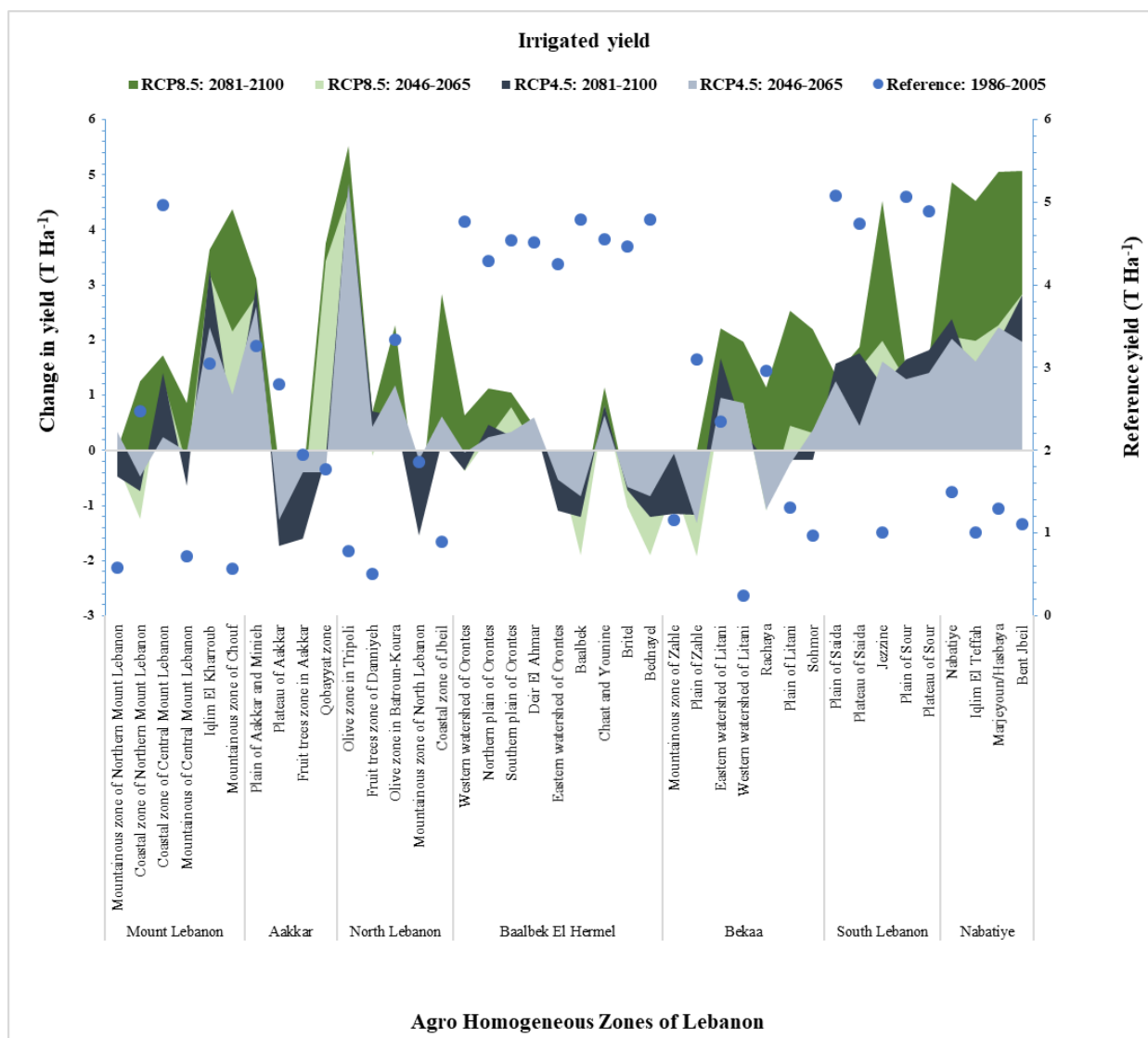


Figure 12 The average irrigated yield for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon

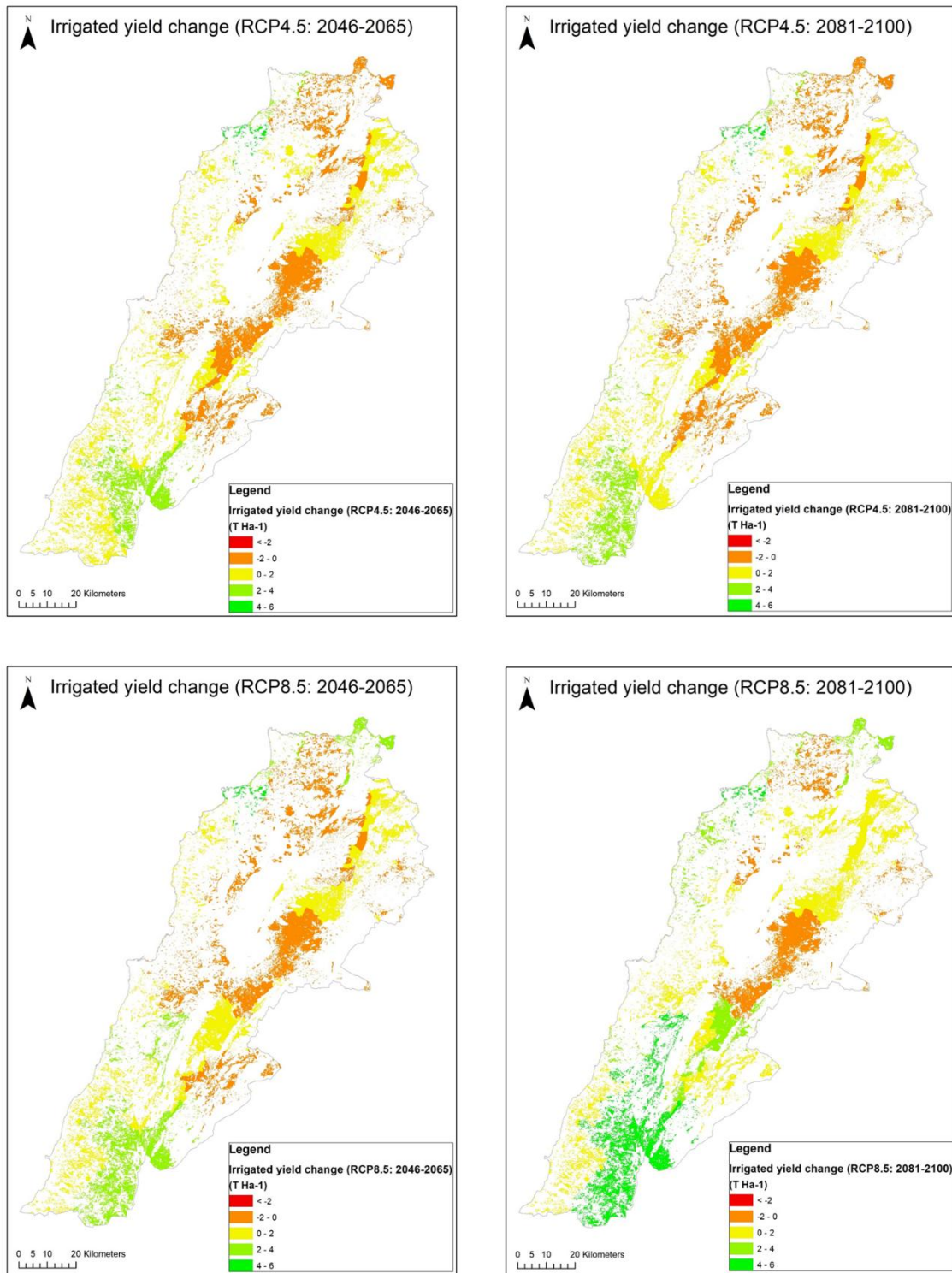


Figure 13 Maps of the future change in irrigated wheat yield under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

3 Impact of climate change on wheat water productivity

Under rainfed conditions, wheat water productivity, being the ratio of yield over the effective rain, is expected to increase in different AHZ under all scenarios. However, in areas, such as Baalbek-Hermel, WP is projected to decrease due to the fact that there will not be enough rain quantities or even too little quantities to trigger planting and yield accumulation.

Under irrigated conditions, the WP is defined as the ratio of yield over the water used from rain and irrigation. It is clear that WP will increase in areas where there will be a decrease in rain amounts and a need to rely more on irrigation. However, at the end-century, under RCP8.5, WP is expected to increase in most of the AHZ of the country which could be explained by the fact that CO₂ fertilization effect will induce an increase in crop yield while the shortening of the growth cycle will play an important role in reducing irrigation amounts, which will be reflected in a general increasing trend of WP.

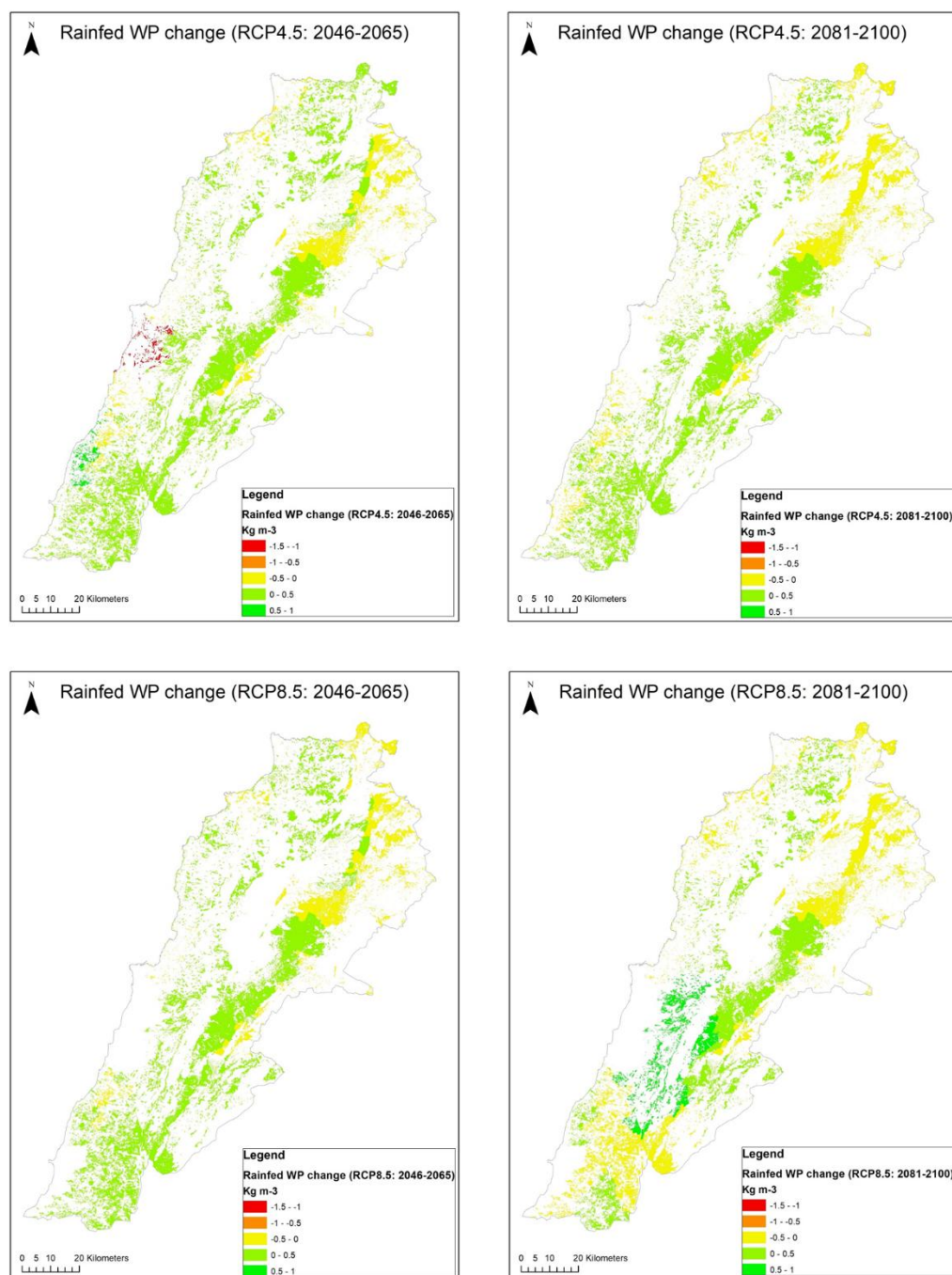


Figure 14 Maps of the future change in rainfed wheat water productivity under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

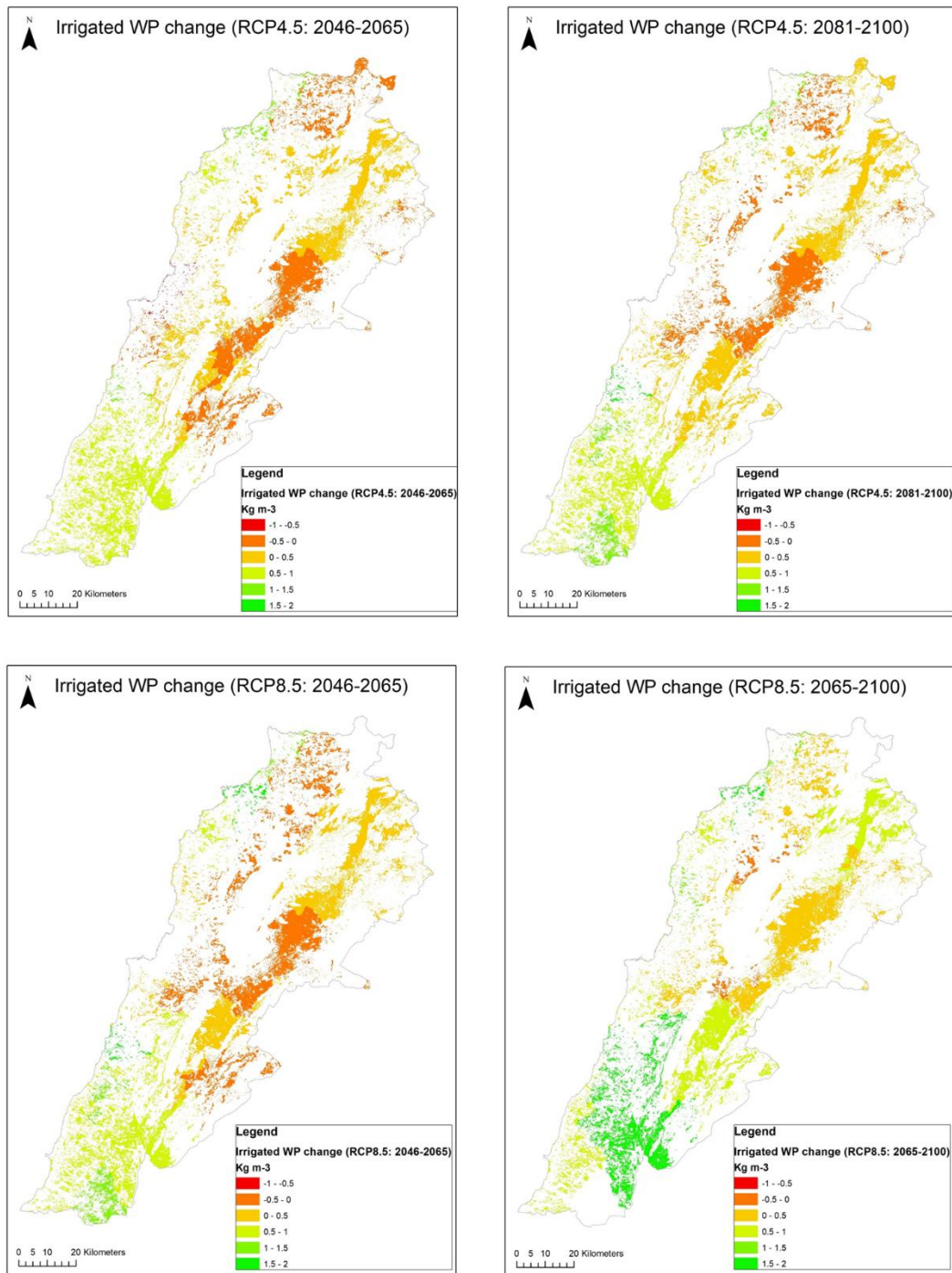


Figure 15 Maps of the future change in irrigated wheat water productivity under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).

Chapter IV: Discussion

In this study we present modelled crop yields for both rainfed and irrigated wheat in the whole of Lebanon. Our rainfed results are in accordance with those done by Bird et al., (2016) that shows that crop yield can decrease in a Mediterranean region (Sardinia) under rainfed conditions due to crop failure. At the same time, the increase in wheat yield in the future is in accordance with the results of Dubey & Sharma (2018) who studied wheat yield changes in Banas river basin, a semi-arid region.

As demonstrated in the research, increasing irrigation is a technological approach that is particularly successful at adjusting to changing climate. However, rather than lowering demand, this technique increases water supplies, which goes against the climatic trend. At some time, there will be insufficient water locally to deliver the necessary irrigation in a sustainable manner. Following that, either a change in infrastructure or the discovery of a new sustainable supply of water will be necessary.

Adjusting planting dates and the use mulches have proven to be highly successful in lowering agricultural value-at-risk due to climate change according to Bird et al., 2016, mulching proved efficient at lowering soil evapotranspiration reducing water loss. Changes in crop types can also be an adaptation strategy tend to increase water-use efficiency and decrease water demand.

With extreme climatic events being on the rise, spring frost or heat waves during growing, flowering, or grain filling season could have a negative impact on yield even in areas where yield will be increasing.

Change of crop varieties, i.e changing from winter wheat to spring wheat varieties could help in areas where crop failure occurs, nevertheless, irrigation waters would still be required in these areas.

Land use change could be an option, but yield is also affected by the type of soil plants are sown on, therefore an assessment will have to be made in order to strategically use available land for wheat production.

Several limitations should be kept in mind when interpreting these results. AquaCrop simulates the response of yield to water, temperature and atmospheric CO₂, but it does not account for losses caused by pests, diseases, weeds, nutrient deficiency or waterlogging, so the simulated values represent an attainable rather than an actual yield. The projected increases are driven to a large extent by the CO₂ fertilization effect, whose real magnitude for field-grown wheat remains uncertain and may be smaller than the model assumes. AquaCrop also represents the effect of short episodes of extreme heat during flowering and grain filling in a simplified way; studies that resolve this stress more explicitly, such as Yang et al. (2019) for rainfed wheat in southern Portugal, project yield losses rather than gains. Furthermore, the studies whose results agree with ours, in particular Dubey & Sharma (2018), used AquaCrop with a rising CO₂ concentration, as we did, so this agreement is not fully independent.

On the climate side, the projections were based on a single regional climate model (RCA4) driven by three global climate models. Relying on one regional model does not capture the

additional uncertainty introduced by the choice of downscaling model, so the range of outcomes shown here is narrower than in multi-model ensemble studies. Finally, the planting date in each simulation was allowed to shift with the onset of the rainy season, which lets the crop partly avoid the driest and hottest part of the year, and the irrigated simulations assume that the required water is available, which may not hold in areas where groundwater is already over-exploited. These results should therefore be read as one plausible trajectory for Lebanese wheat under a changing climate rather than as a precise forecast.

Chapter V: Conclusion and future work

In this paper, we have shown that temperatures are expected to increase while precipitations will decrease. Under irrigated conditions, wheat yield will increase in the future in comparison to the reference period (1986-2005) in most AHZs in Lebanon with highest increases being in Mount Lebanon, South and Nabatiyeh in the far future (2081-2100) in the RCP8.5 projection. The smallest increase will be in the Baalbek-Hermel region where warm temperature will stop plants from fully developing reducing yield even with available irrigation. Water productivity will also increase due to the shortening of the growing season of the crops. In order to avoid the depletion of water resources, sustainability studies could be conducted to find more ways to mitigate the climate change impact on crop yields.

References

- 2° Institute. (2020a). *Methane Levels: Current & Historic Atmospheric CH₄*. Retrieved from: <https://www.methanelevels.org/#sources>. On: Tuesday, April 27, 2021.
- 2° Institute. (2020b). *Nitrous Oxide Levels: Current & Historic Atmospheric N₂O*. Retrieved from: <https://www.n2olevels.org/>. On: Tuesday, April 27, 2021.
- Abd-Elmabod, S. K., Muñoz-Rojas, M., Jordán, A., Anaya-Romero, M., Phillips, J. D., Jones, L., Zhang, Z., Pereira, P., Fleskens, L., van der Ploeg, M., & La Rosa, D. de (2020). Climate change impacts on agricultural suitability and yield reduction in a Mediterranean region. *Geoderma*, 374, 114453. <https://doi.org/10.1016/j.geoderma.2020.114453>
- Abi Saab, M. T., El Alam, R., Jomaa, I., Skaf, S., Fahed, S., Albrizio, R., & Todorovic, M. (2021). Coupling Remote Sensing Data and AquaCrop Model for Simulation of Winter Wheat Growth under Rainfed and Irrigated Conditions in a Mediterranean Environment. *Agronomy*, 11(11), 2265. <https://doi.org/10.3390/agronomy11112265>
- Abi Saab, M. T., Houssemeddine Sellami, M., Giorio, P., Basile, A., Bonfante, A., Rouphael, Y., Fahed, S., Jomaa, I., Stephan, C., Kabalan, R., Massaad, R., Todorovic, M., & Albrizio, R. (2019). Assessing the Potential of Cereal Production Systems to Adapt to Contrasting Weather Conditions in the Mediterranean Region. *Agronomy*, 9(7), 393. <https://doi.org/10.3390/agronomy9070393>

- Abi Saab, M. T., Jomaa, I., Skaf, S., Fahed, S., & Todorovic, M. (2019). Assessment of a Smartphone Application for Real-Time Irrigation Scheduling in Mediterranean Environments. *Water*, 11(2), 252. <https://doi.org/10.3390/w11020252>
- Arab Center for the Studies of Arid Zones and Dry Lands (ACSAD) of the League of Arab States, Ministry of Agriculture in Lebanon, National Council for Scientific Research, & United Nations Economic and Social Commission for Western Asia. (2019). *Integrated Vulnerability Assessment Application on the Lebanese Agricultural Sector*. United Nations Publication. Retrieved from: http://riccar.org/index.php/publications/integrated-vulnerability-assessment-application-lebanese-agricultural-sector?language_content_entity=en.
- Bahri, H., Annabi, M., Cheikh M'Hamed, H., & Frija, A. (2019). Assessing the long-term impact of conservation agriculture on wheat-based systems in Tunisia using APSIM simulations under a climate change context. *The Science of the Total Environment*, 692, 1223–1233. <https://doi.org/10.1016/j.scitotenv.2019.07.307>
- Batisani, N., & Yarnal, B. (2010). Rainfall variability and trends in semi-arid Botswana: Implications for climate change adaptation policy. *Applied Geography*, 30(4), 483–489. <https://doi.org/10.1016/j.apgeog.2009.10.007>
- Bird, D. N., Benabdallah, S., Gouda, N., Hummel, F., Koeberl, J., La Jeunesse, I., Meyer, S., Prettenthaler, F., Soddu, A., & Woess-Gallasch, S. (2016). Modelling climate change impacts on and adaptation strategies for agriculture in Sardinia and Tunisia using AquaCrop and value-at-risk. *The Science of the Total Environment*, 543(Pt B), 1019–1027. <https://doi.org/10.1016/j.scitotenv.2015.07.035>
- Blanco, M., Ramos, F., van Doorslaer, B., Martínez, P., Fumagalli, D., Ceglar, A., & Fernández, F. J. (2017). Climate change impacts on EU agriculture: A regionalized perspective taking into account market-driven adjustments. *Agricultural Systems*, 156, 52–66. <https://doi.org/10.1016/j.agsy.2017.05.013>
- Bocchiola, D., Brunetti, L., Soncini, A., Polinelli, F., & Gianinetto, M. (2019). Impact of climate change on agricultural productivity and food security in the Himalayas: A case study in Nepal. *Agricultural Systems*, 171, 113–125. <https://doi.org/10.1016/j.agsy.2019.01.008>
- Darwish et al. (2006). Soil Map of Lebanon.
- Dubey, S. K., & Sharma, D. (2018). Assessment of climate change impact on yield of major crops in the Banas River Basin, India. *The Science of the Total Environment*, 635, 10–19. <https://doi.org/10.1016/j.scitotenv.2018.03.343>
- FAO (2006). Agricultural Homogeneous Zones: Support to the Agricultural Census, 234.
- Hassanli, M., Ebrahimian, H., Mohammadi, E., Rahimi, A., & Shokouhi, A. (2016). Simulating maize yields when irrigating with saline water, using the AquaCrop, SALTMED, and SWAP models. *Agricultural Water Management*, 176, 91–99. <https://doi.org/10.1016/j.agwat.2016.05.003>

- IPCC (2014). Climate Change 2014: Synthesis Report. Contribution. 0309-0167. Retrieved from: <https://www.ipcc.ch/report/ar5/wg2/>.
- IPCC, Mbow, C., Rosenzweig, C., Barioni, L. G., Benton, T. G., Shukla, P. R., Skea, J., Calvo Buendia, E., Masson-Delmotte, V., Pörtner, H.-O., Roberts, D. C., Zhai, P., Slade, R., Connors, S., van Diemen, R., Ferrat, M., Haughey, E., Luz, S., Neogi, S.,... Malley, J. (2021). *Special Report on Climate Change And Land - Chapter 5: Food security*. Retrieved from: <https://www.ipcc.ch/srccl/chapter/chapter-5/>
https://www.ipcc.ch/site/assets/uploads/sites/4/2021/02/08_Chapter-5_3.pdf.
- Kim, C.-G. (2008). The Impact of Climate Change on the Agricultural Sector: Implications of the Agro-Industry for Low Carbon, Green Growth Strategy and Roadmap for the East Asian Region. *Korea Rural Economic Institute*.
- Kukal, M. S., & Irmak, S. (2018). Climate-Driven Crop Yield and Yield Variability and Climate Change Impacts on the U.S. Great Plains Agricultural Production. *Scientific Reports*, 8(1), 3450. <https://doi.org/10.1038/s41598-018-21848-2>
- Lesk, C., Rowhani, P., & Ramankutty, N. (2016). Influence of extreme weather disasters on global crop production. *Nature*, 529(7584), 84–87. <https://doi.org/10.1038/nature16467>
- Lindsey, R., & Dahlman, L. (2021). *Climate Change: Global Temperature | NOAA Climate.gov*. Retrieved from: <https://www.climate.gov/news-features/understanding-climate/climate-change-global-temperature>. On: Friday, May 14, 2021.
- MoA, & FAO (2010). Agricultural Census 2010.
- Muluneh, A. (2020). Impact of climate change on soil water balance, maize production, and potential adaptation measures in the Rift Valley drylands of Ethiopia. *Journal of Arid Environments*, 179, 104195. <https://doi.org/10.1016/j.jaridenv.2020.104195>
- NASA. (2019). *Evidence | Facts – Climate Change: Vital Signs of the Planet*. Retrieved from: <https://climate.nasa.gov/evidence/>. On: Thursday, May 20, 2021.
- NASA. (2021). *Effects | Facts – Climate Change: Vital Signs of the Planet*. Retrieved from: <https://climate.nasa.gov/effects/>. On: Tuesday, May 11, 2021.
- Piervitali, E., & Colacino, M. (2003). Precipitation Scenarios in the Central-western Mediterranean Basin. In *Mediterranean Climate* (pp. 245–258). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-55657-9_14
- Prăvălie, R., Sîrodoev, I., Patriche, C., Roșca, B., Piticar, A., Bandoc, G., Sfîcă, L., Tișcovschi, A., Dumitrașcu, M., Chifiriuc, C., Mănoiu, V., & Iordache, Ș. (2020). The impact of climate change on agricultural productivity in Romania. A country-scale assessment based on the relationship between climatic water balance and maize yields in recent decades. *Agricultural Systems*, 179. <https://doi.org/10.1016/j.agsy.2019.102767>
- Saxton, K. E., Rawls, W. J., Romberger, J. S., & Papendick, R. I. (1986). Estimating Generalized Soil-water Characteristics from Texture. *Soil Science Society of America Journal*, 50(4), 1031–1036. <https://doi.org/10.2136/sssaj1986.03615995005000040039x>

- Steduto, P., Hsiao, T. C., Fereres, E., & Raes, D. (2012). *Crop yield response to water*. FAO irrigation and drainage paper, 0254-5284: Vol. 66. FAO. Retrieved from: <http://www.fao.org/docrep/016/i2800e/i2800e.pdf>.
- Tahmasebi, T., Karami, E., & Keshavarz, M. (2020). Agricultural land use change under climate variability and change: Drivers and impacts. *Journal of Arid Environments*, 180, 104202. <https://doi.org/10.1016/j.jaridenv.2020.104202>
- Verner, D., Ashwill, M., Christensen, J., McDonnell, R., Redwood, J., Jomaa, I., Saade, M., Massad, R., Chehade, A., Bitar, A., & Treguer, D. (2018). *Droughts and Agriculture in Lebanon*. World Bank, Washington, DC. Retrieved from: <https://openknowledge.worldbank.org/handle/10986/30595>. <https://doi.org/10.1596/30595>
- Voloudakis, D., Karamanos, A., Economou, G., Kalivas, D., Vahamidis, P., Kotoulas, V., Kapsomenakis, J., & Zerefos, C. (2015). Prediction of climate change impacts on cotton yields in Greece under eight climatic models using the AquaCrop crop simulation model and discriminant function analysis. *Agricultural Water Management*, 147, 116–128. <https://doi.org/10.1016/j.agwat.2014.07.028>
- Warnatzsch, E. A., & Reay, D. S. (2020). Assessing climate change projections and impacts on Central Malawi's maize yield: The risk of maladaptation. *The Science of the Total Environment*, 711, 134845. <https://doi.org/10.1016/j.scitotenv.2019.134845>
- Yang, C., Fraga, H., van Ieperen, W., Trindade, H., & Santos, J. A. (2019). Effects of climate change and adaptation options on winter wheat yield under rainfed Mediterranean conditions in southern Portugal. *Climatic Change*, 154(1–2), 159–178. <https://doi.org/10.1007/s10584-019-02419-4>
- Zhu, X., Xu, K., Liu, Y., Guo, R., & Chen, L. (2021). Assessing the vulnerability and risk of maize to drought in China based on the AquaCrop model. *Agricultural Systems*, 189, 103040. <https://doi.org/10.1016/j.agsy.2020.103040>

LIST OF ABBREVIATIONS

1. NOAA: National Oceanic and Atmospheric Administration.
2. NASA: National Aeronautics and Space Administration, U.S.A.
3. FAO: The Food and Agriculture Organization of the United Nations.
4. LARI: Lebanese Agricultural Research Institute.
5. AHZ: Agro-Homogeneous Zone.
6. CC: Climate change.
7. CO₂: Carbon dioxide.
8. N₂O: Nitrous oxide.
9. GHG: Greenhouse gas.
10. IPCC: Intergovernmental Panel on Climate Change.
11. RCP: Representative Concentration Pathway.
12. GCM: General Circulation Model.

13. RCM: Regional Climate Model.
14. CORDEX: Coordinated Regional Climate Downscaling Experiment.
15. EURO-CORDEX: European domain of CORDEX.
16. RCA4: Rossby Centre regional Atmospheric model, version 4.
17. SMHI: Swedish Meteorological and Hydrological Institute.
18. ESGF: Earth System Grid Federation.
19. GIS: Geographic Information System.
20. QGIS: Quantum GIS (open-source GIS software).
21. netCDF: Network Common Data Form.
22. MoA: Ministry of Agriculture (Lebanon).
23. ACSAD: Arab Center for the Studies of Arid Zones and Dry Lands.
24. SDG: Sustainable Development Goal.
25. WP: Water productivity.
26. ET: Evapotranspiration.
27. ET0: Reference evapotranspiration.
28. GDD: Growing degree days.
29. LAI: Leaf area index.
30. HI: Harvest index.
31. ppm: parts per million.
32. ppb: parts per billion.

LIST OF SYMBOLS

- B: cumulative aboveground biomass (kg/m²).
- WP: water productivity parameter (kg/m² per mm, or kg per m³ of water transpired).
- Tr: crop transpiration (mm).
- Y_x: potential (maximum) crop yield (kg/m²).
- Y₀: actual crop yield (kg/m²).
- ET_x: potential evapotranspiration (mm).
- ET₀: actual evapotranspiration (mm).
- K_y: yield response factor (crop yield to water-stress coefficient).
- HI: harvest index.

LIST OF FIGURES

- Figure 1 Global average surface temperature change7
- Figure 2 Annual anthropogenic CO₂ emissions according to different RCPs.....12
- Figure 3 The Agro Homogeneous Zones of Lebanon and potential wheat growing area.....18
- Figure 4 The average rain for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon22

Figure 5 Maps of the future change in rain over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	23
Figure 6 The average minimum temperature for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon.	25
Figure 7 Maps of the future change in minimum temperature over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	26
Figure 8 The average maximum temperature for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon	27
Figure 9 Maps of the future change in maximum temperature over Lebanon under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	28
Figure 10 The average rainfed yield for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon.	29
Figure 11 Maps of the future change in rainfed wheat yield under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	30
Figure 12 The average irrigated yield for the reference period (1986-2005) and corresponding future changes under RCP4.5 and RCP8.5 for the mid-century (2046-2065) and end-century (2081-2100) for the different AHZ of Lebanon	32
Figure 13 Maps of the future change in irrigated wheat yield under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	33
Figure 14 Maps of the future change in rainfed wheat water productivity under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	35
Figure 15 Maps of the future change in irrigated wheat water productivity under scenarios RCP4.5 and 8.5 for the mid-century (2046-2065) and end-century (2081-2100) in respect to the reference period (1986-2005).	36

LIST OF TABLES

Table 1 Results from different crop simulation models	14
Table 2 Example of AquaCrop use for climate change simulations.	16
Table 3 The description of the different Agro Homogeneous Zones of Lebanon	18